

APPROXIMATING THE EQUILIBRIUM EFFECTS OF INFORMED SCHOOL CHOICE

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We study how information about school quality affects families' choices and how scaling up the intervention changes its impact. In a randomized trial in Chile, we offer families a short video and a personalized school report card before they enroll their children. Treated families choose schools with higher test scores and value added. Their children score about 0.2 standard deviations higher in fourth grade, with positive estimates in later observed assessment samples. We embed the experimentally identified demand shift in a structural model of school choice and competition to study equilibrium effects at scale. Capacity constraints and centralized assignment attenuate the demand-side gains, but the same shift in demand strengthens schools' incentives to improve quality, and the supply response largely offsets the congestion effect. More generally, equilibrium scaling can alter experimental effects in either direction because congestion and endogenous quality responses work against each other. We use the model to guide the design of a follow-up cluster-randomized experiment, focusing on the tradeoff between statistical power and equilibrium spillovers.

KEYWORDS: Information, Incentives, School Choice, Supply Side, Randomized Controlled Trial.

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1. INTRODUCTION

A growing body of evidence shows that providing families with information about school quality can shift their school choices (Hastings and Weinstein, 2008, Andrabi et al., 2017, Kapor et al., 2020, Campos, 2024, Corradini, 2024). But what happens when such a policy is implemented at scale? In a small-scale experiment, school characteristics are, to a first approximation, fixed, and each treated family's choice has negligible effects on market conditions. At scale, this approximation breaks down. Information can concentrate demand at particular schools, causing families to compete for a limited number of seats and changing schools' incentives to invest in quality. A small-scale experiment therefore identifies the direct choice response, but not the congestion and supply responses that emerge at scale. Whether these forces reinforce or undo the experimental gains determines what the small-scale experiment implies for implementing the information intervention more broadly.²

This paper studies these equilibrium effects in Chile, a well-studied setting with extensive school choice and a large private voucher sector. To do so, we first designed a school-choice information intervention with scale in mind, using a low-cost format that could be delivered through Chile's existing public preschool infrastructure. The intervention combines a short video with a personalized school report card: the video emphasizes the link between school quality and future educational and labor-market opportunities, while the report card compares nearby primary schools on test scores, prices, and location and includes a map. We examine the intervention's direct effects on school choice in a small-scale randomized controlled trial. We find that treated families choose schools with higher test scores and value added. These effects are concentrated among families reached before they had enrolled their child in a primary school, when the intervention could still influence the initial choice. These experimentally identified choice effects provide the demand-side inputs to the structural model. Separately, we follow children in administrative data to assess whether the changes in school choice were consequential for learning. Among families reached before enrollment, treated children score about 0.2 standard deviations higher on fourth-grade assessments. Point estimates remain positive in eighth grade and on the college entrance examination more than a decade later, but these

²The distinction between partial and equilibrium effects has long been recognized as important in education contexts (Heckman et al., 1998). Andrabi et al. (2017) show that information can induce equilibrium responses, including supply-side adjustment, in small village schooling markets in Pakistan. Direct evidence for large urban markets, however, remains limited.

later assessments are observed for smaller and potentially selected test-taking samples, so we interpret them as supporting evidence rather than as a separate long-run finding.

To move from the experiment to an equilibrium analysis at scale, we build on the demand model estimated by Neilson (2026).³ The baseline demand system features heterogeneous families choosing among schools based on quality, out-of-pocket price, and distance across 53 urban markets. Following the scaled-logit logic in Brown and Jeon (2024), we extend this system to allow the information intervention both to raise the weight on school characteristics targeted by the report cards and to make choices more responsive to school characteristics generally. We estimate the parameters governing these two responses by matching the experimental effects on chosen-school characteristics to their model-simulated counterparts using GMM. Once estimated, these parameters describe how information changes the way families evaluate the options they face. The extrapolation then uses the full structure of each family’s local choice problem. Observationally similar families may live near different schools, receive report cards describing different alternatives, and respond differently because the quality, prices, and distances in their choice sets differ. Combining the experimentally identified behavioral response with these heterogeneous local environments allows the model to map the experiment to the broader population.

The change in demand creates the supply-side mechanism at the heart of the paper. By making families more responsive to quality, information can compress schools’ quality markdowns and strengthen their incentives to invest. To quantify this response, we take as our starting point the quality first-order condition derived by Neilson (2026), which relates a school’s quality choice to marginal revenue, demand responsiveness, and costs. Neilson (2026) uses this condition to interpret schools’ historical response to Chile’s SEP reform (*Subvención Escolar Preferencial*), which raised the voucher for low-income students. We take this intuition one step further by estimating an empirical supply model on a ten-year administrative panel and extending the quality condition to allow for capacity constraints. The estimated cost parameters allow us to predict counterfactual quality choices. Unconstrained schools can respond to stronger demand by expanding enrollment, whereas schools near capacity can respond through student composition, including by attracting students eligible for the higher SEP voucher. Identification combines reform-induced variation in school revenue with local labor-market shifters, includ-

³Related work on school choice and competition in education markets includes Neilson (2026), Dinerstein and Smith (2015), Walters (2018), Kapor et al. (2020). The model in this paper also connects to earlier work on school choice in Chile by Gallego and Hernando (2008) and Chumacero et al. (2011).

ing non-education earnings and a new inverse-distance-weighted index of nearby public-school teacher quality. In the counterfactual, centralized assignment determines which schools become constrained, creating congestion on the demand side, while the supply model determines how schools adjust quality in response to the change in competitive incentives.

We build the counterfactual in three steps. The first is a demand-only benchmark that holds school characteristics fixed and ignores capacity constraints. We apply the experimentally identified change in demand to the target population while preserving each family's socioeconomic characteristics, location, and local school options, translating the experimental results into the effects of a broader intervention in the environments where the policy would operate. The resulting population average need not equal the experimental-sample estimate because families in the two populations face different distributions of school quality, prices, and distance. The detailed spatial data also allow the model to recover the distribution of effects across families and locations, rather than only an average effect. In this benchmark, the demand shift raises the expected quality (value added) of the schools attended by low-income families by 0.186 SD. Adding capacity constraints attenuates the effect by roughly half because of congestion at popular schools (from 0.186 to 0.109 SD). Allowing incumbent schools to adjust quality, while holding entry, marginal costs, and capacity fixed, captures a medium-run supply response that more than offsets this attenuation. The final equilibrium effect is 0.229 SD, slightly above but not statistically distinguishable from the demand-only benchmark. The mechanism behind this reversal is that more quality-responsive demand raises schools' returns to improving quality. More generally, scaling a demand-shifting policy in a market with finite capacity and endogenous quality activates two opposing forces: congestion reallocates access to existing quality, while stronger competitive incentives can induce providers to produce more of it. Their relative magnitude is an empirical property of the market, so the direction of the equilibrium correction cannot be inferred from the small-scale experimental effect alone.

We make three contributions to the literature. First, we provide experimental evidence on families' responses to school-choice information and study the policy's equilibrium consequences in a large urban market. Andrabi et al. (2017) provide the closest precedent, showing that school report cards in Pakistan's small village markets raised achievement and induced supply-side responses. In that setting, report cards reach entire communities, so market-level variation arises naturally. Large urban markets, like those we study in Chile, pose a different challenge: schools draw from spatially dispersed populations and families face many alternatives, so demand shifts diffuse across a wide geography rather than concentrating in a few

schools. An experimental design that saturates some urban markets while preserving others as credible control groups may not be feasible, which is why the structural model is needed to study the policy at scale. The counterfactual results show that both congestion and schools' quality responses are quantitatively important for the policy assessment.

Second, we show how a demand shift identified in a small-scale experiment can discipline a prospective equilibrium policy evaluation. Our approach lies between two familiar uses of structural models. In one, the policy has already been implemented at scale and the model decomposes observed effects (e.g., Neilson, 2026, Dinerstein and Smith, 2015). In the other, the entire counterfactual rests on the model, with no experimental variation to pin down the demand response. Here, the experiment identifies a credible demand shift that is observed in the data, but the full policy environment—congestion, assignment constraints, supply-side responses—must be studied through the model, because these forces emerge only when the treated population is large enough to change market conditions. A comparison with Neilson (2026) makes the distinction concrete: that paper studies Chile's SEP voucher reform after it was implemented at scale, and uses the model to recover demand and decompose the observed equilibrium response into its underlying competitive forces. For the information policy studied here, no such implementation exists, so the model must recover cost parameters, introduce capacity constraints, and solve for the counterfactual quality equilibrium. In this sense, the model is used for prospective policy evaluation rather than retrospective decomposition, and the demand shift rests on the experiment rather than on the model alone.⁴

Third, we show how the estimated model can inform the design of a larger-scale experiment that would detect these equilibrium effects directly rather than through the model. One middle ground between randomizing entire markets and randomizing individual families is to define smaller geographic clusters within markets and assign them independently to treatment or control. The challenge, however, is that when a policy generates equilibrium responses that propagate through shared schools, treated and control clusters are not independent, and the trade-off between cluster size and statistical power depends on the spatial reach of these spillovers. In Section 7, we use the model to navigate this trade-off, studying how cluster size, buffer zones, and take-up rates shape the bias and precision of a cluster-randomized trial in Santiago. The

⁴Todd and Wolpin (2023) survey the literature combining experiments with structural models for policy evaluation, including cases where experiments estimate or validate structural parameters (Todd and Wolpin, 2006, Walters, 2018). A related approach to ours appears in Lise et al. (2015), who combine experimental variation with equilibrium considerations in the context of employment.

analysis reveals that the design problem depends on which equilibrium channels are active: the trade-off between contamination and power is not binding when only the demand response must be detected, but it binds once the target is the supply-side quality response, which propagates across cluster boundaries through competition among nearby schools.

Taken together, the paper offers a way to assess demand-shifting policies whose broad implementation could change both access and supply. The framework separates three objects: the behavioral response identified by the experiment, the distribution of families and the alternatives they face in the target population, and the equilibrium responses that emerge when the intervention is scaled. The broader value of this approach extends beyond Chile: it provides a disciplined way to evaluate policies when an experiment identifies individual responses, but broad implementation changes both access and provider incentives.

The rest of the paper is organized as follows. Section 2 provides a brief conceptual overview of the economic forces that motivate the analysis. Section 3 describes the institutional setting in Chile, the design of the information intervention, and the data. Section 4 presents the experimental results. Section 5 develops the structural model and reports parameter estimates. Section 6 simulates the equilibrium effects of scaling up the information policy. Section 7 uses the model to design a larger-scale experiment. Section 8 concludes.

2. CONCEPTUAL FRAMEWORK

2.1. *Framework for Policy Analysis*

This section summarizes the policy problem that motivates the empirical analysis. The randomized controlled trial identifies how the intervention changes school choices for the families we study. A policymaker, however, is interested in a different question: what would happen if the same intervention were implemented more broadly, in places where families face different local school environments and where the intervention itself may change market conditions? The experiment identifies the demand shift induced by the policy; the challenge for the rest of the paper is to understand how that shift propagates once the policy is scaled up. This is the sense in which the paper combines experimental and structural evidence: the experiment provides the demand shift, and the model provides the equilibrium analysis that the experiment alone cannot deliver.

Two features of this setting make that extrapolation nontrivial. First, school choice is highly localized. Consider two families with the same socioeconomic characteristics and voucher eligibility but living in different neighborhoods. Both are assigned to the same intervention, yet

their personalized report cards describe different nearby schools. One family may learn about substantial quality differences among schools available at no cost, while the other may face nearby schools with similar quality or better schools only at considerably greater distance. The information families receive, and their scope to respond to it, therefore depend on the interaction between their characteristics and the local school environment.

Experimental heterogeneity analysis by socioeconomic characteristics can reveal which types of families respond more to the treatment, but it cannot capture this interaction in full. The relevant environment comprises the quality, prices, and distances associated with all the schools a family might attend—a multidimensional spatial object that cannot generally be reduced to a single subgroup characteristic. We observe these configurations in the data, and the structural model applies the experimentally identified change in choice behavior to the particular environment each family faces. The treatment effect measured in the experimental sample therefore need not carry mechanically to a broader population with a different spatial and socioeconomic composition.

Second, scaling the intervention can generate equilibrium effects. When only a small group of families is treated, school quality, assignment probabilities, and school incentives are, to a first approximation, fixed. When many families are treated, that approximation becomes less credible. Increased demand for higher-quality schools can create congestion, especially when access is mediated by assignment and school capacity. In the medium run, changes in demand can also alter schools' incentives to improve quality. The aggregate effect of the policy may therefore differ from the partial-equilibrium effect measured in a small-scale experiment.

The empirical strategy in the rest of the paper is designed to address these two issues jointly. We combine the experimental variation with a model of school choice that allows treatment effects to vary with local choice environments, then embed that demand system in counterfactual exercises with assignment, finite capacity, and school quality responses. This provides a disciplined way to move from the experimentally identified effect for the treated sample to policy-relevant predictions for larger populations. In particular, it lets us study three objects that a small-scale experiment alone cannot recover: how the policy transports across different local choice environments, how congestion alters assignments when many families are treated, and how schools adjust quality in response to the resulting competitive pressure.

Beyond these policy counterfactuals, the framework also informs the design of larger-scale experiments. Because the model quantifies how demand shifts propagate spatially and how supply-side responses spill across geographic boundaries, it provides the ingredients needed to

evaluate candidate cluster-randomized designs before they are fielded. We apply this logic in Section 7, where we design a trial in Santiago and show that the optimal configuration depends on which equilibrium channels are active.

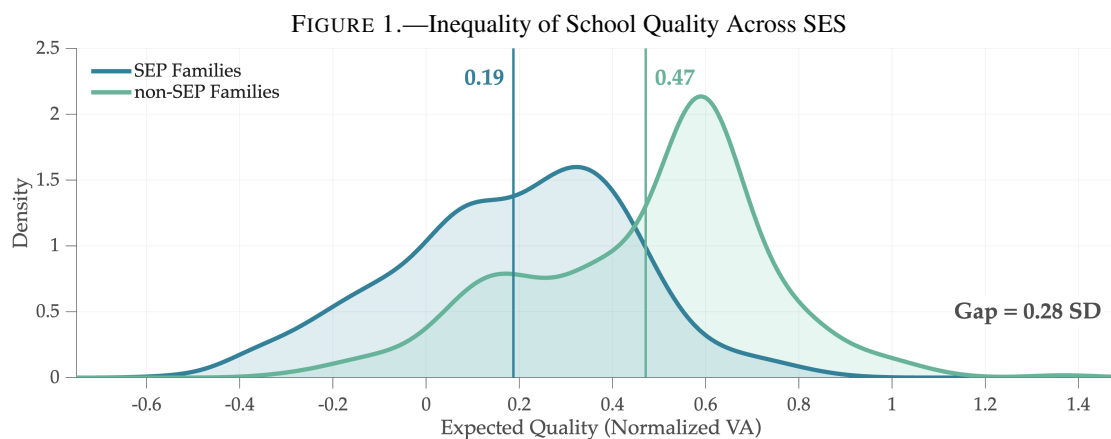
3. SETTING, INTERVENTION, AND DATA

3.1. *Institutional Setting*

Primary schools in Chile are either free public schools, private voucher schools, or private non-voucher schools. The system features a high degree of school choice and a large private sector: in 2016, the market share for first-grade students was 36% for public schools, 55% for voucher schools, and 8% for private schools. Voucher schools may charge copayments, but these are capped and interact with targeted subsidies for disadvantaged students. In particular, in 2008 the government introduced the SEP reform (*Subvención Escolar Preferencial*), which increased the voucher for the poorest 40% of students. Participating schools were required to charge zero tuition to eligible families and were prohibited from screening applicants, including on academic grounds. The 2009 General Education Law extended the prohibition on selection more broadly across subsidized elementary schools. The 2015 Inclusion Law subsequently froze existing copayments and began replacing them with public funding. Chile later replaced school-by-school applications with a centralized application system that lets families rank multiple schools in a common process. When a school is oversubscribed, assignments follow centrally specified priorities and lottery numbers. Centralization therefore makes applications and the allocation of scarce seats more transparent; it did not introduce the earlier restriction on school selection.⁵

In spite of the variety of schools and choices available, students from poorer families tend to attend schools with lower academic outcomes and lower inputs. Figure 1 shows model-predicted baseline expected school quality for the structural experiment sample by SEP eligibility, which proxies for socioeconomic status. One reason for this gap is that poorer families may place less weight on academic quality when choosing schools, either because quality is harder to observe or because the returns to choosing a better school are less salient. Chilean policymakers have long published school performance data, but this information is rarely tailored to the schools in a family's immediate vicinity or timed to reach parents when they are actively choosing where to enroll. Our intervention targets both gaps.

⁵See Neilson (2026) for a detailed description of the market structure and policy environment.



Notes: Distribution of model-predicted baseline expected school quality (normalized VA) for the 815 families in the structural experiment sample, evaluated in their 2011 model choice environments. Each family receives equal weight within its SEP group. SEP families are eligible for Chile’s targeted voucher (*Subvención Escolar Preferencial*), which proxies for low socioeconomic status. Vertical lines mark group means.

We reach families through the public preschool system. Pre-K attendance is widespread before primary school entry, with net enrollment rates of 55% at age 3 and 87% at age 4 (OECD, 2016). We partnered with *Fundación Integra*, a national network of approximately 1,000 public preschool centers serving low-income families throughout the country. We delivered the intervention at preschool parent meetings, reaching families during the period when they are actively deciding which primary school to enroll their child in for the following year. At the time of our experiment (2010), each school managed its own admissions and timing varied across schools. Chile has since adopted a centralized application system with a common enrollment window, which would make the timing of information provision more predictable in a scaled-up policy environment. The Online Appendix provides additional institutional detail on preschool provision.

3.2. Information Intervention and Experimental Design

The intervention was designed to make school choice more salient and to provide parents with simple, comparable information about nearby schools. Its goal was to increase both awareness of local school characteristics and the perceived returns to school quality. The treatment was delivered at regular *Integra* parent-teacher meetings and combined a short video with a personalized report card. The video emphasized that choosing a school carefully can shape a child’s future opportunities: it linked school quality to preparation for higher education and better jobs, and featured role models from low-income neighborhoods who had accessed better opportuni-

ties through education (Figure O-2 and Figure O-3). Each family also received a report card listing test scores, prices, and locations of nearby primary schools, together with a map to help parents locate and compare options (Figure O-1).

We implemented the study in the three largest regions of Chile: Valparaíso, Biobío, and Santiago. Preschools were eligible if they were in urban areas, had at least 10 primary schools within 2 km, and were located in municipalities with at least two primary schools per preschool. These criteria were dictated by the intervention: each family received a personalized comparison of nearby schools, so the experiment required a setting with enough local alternatives for the report card to present a meaningful choice. The criteria yielded 143 eligible preschools, which we randomly assigned to treatment and control before arranging visits, stratifying by region, the number of grades offered, and a measure of local competition given by the number of primary schools within 2 km. Ten could not be visited because of principal refusal, scheduling conflicts, or low expected attendance, leaving 133 participating preschools.⁶ The original design distinguished between two treatment intensities, but imperfect tracking during implementation prevents us from separating the two arms. We therefore pool both versions and interpret the estimates as intent-to-treat effects of assignment to the bundled information package. Additional details on sample selection, including a comparison of participating and nonparticipating eligible centers, treatment materials, and implementation appear in Online Appendix Sections O-2 and O-3. Figure O-5 maps the spatial distribution of treatment and control preschools and student homes in Santiago, and Figure O-6 shows how treated students are distributed across destination schools.

The experiment was implemented between August and December 2010 by trained staff who attended preschool parent meetings. In the 133 preschools that agreed to participate, 1,832 parents signed informed consent forms and completed a baseline survey before any information was provided. The baseline survey collected contact information and information on the application process, including whether the parent expected the child to enter primary school in 2011, whether a school had already been chosen, and whether the family had older children already attending primary school. Parents in treatment preschools then received the intervention. Because our presence at the meeting was not announced in advance, parents could not

⁶As a later benchmark for the size of the relevant network, the official 2018 preschool census lists 257 urban Integra centers in the same 41 municipalities represented by the experimental preschools. The 2018 count is not an exact denominator for the 2010 sampling frame because centers may have opened, closed, moved, or changed identifiers over the intervening eight years.

attend in anticipation of receiving the intervention. Between May and July 2011 we conducted a follow-up survey on enrollment decisions and successfully re-interviewed 1,612 of the 1,832 baseline respondents (88%). We also matched 1,795 of the 1,832 parents in the original sample to administrative records using individual student identifiers (98%).

3.3. *Data, Outcomes, and Analysis Samples*

We use several complementary data sources. First, we use administrative data on preschools from *Integra*, including location, enrollment, attendance, and socioeconomic composition. Second, we use self-collected data from the baseline and follow-up surveys in the experimental preschools. These data include contact information, individual identifiers, family location, and questions regarding the application process. We match this information to administrative data from Chile’s Ministry of Education.

The Ministry of Education data include student-by-year matriculation records, information on grade progression and demographics, eligibility for the SEP voucher, and standardized test scores from SIMCE. We complement these data with birth records from the Ministry of Health, which contain information on child health at birth and on parental background, and with administrative records on all schools in the country, including school type, enrollment, address, tuition, and other characteristics. For the supply analysis, we also use a school-year panel with transfers, expenditures, and measures of teacher and principal quality.⁷

Different parts of the paper use different slices of these data. The reduced-form analysis uses the experimental cohort treated in 2010 and their observed first-grade choices in 2011. We take the baseline demand preference parameters and school–market–year mean utilities from Neilson (2026), who estimates them using administrative records on the full population of first-grade choices and school market shares in 53 urban markets in 2006, 2007, 2011, and 2012. The *Integra* eligibility rules therefore do not enter the market-share inversion. Our second-step treatment estimation uses the subset of experimental families who had not yet enrolled and can be matched to the six urban markets shared with the baseline demand model. Holding the baseline school attributes and mean utilities fixed, this step estimates how the intervention changes the weights families place on observed school characteristics and the scale of choice behavior so as to match the experimental changes in chosen-school characteristics. The supply estimation uses the same 53-market urban sample extended to a 2006–2015 school-year panel,

⁷For further details, see the Online Appendix and Neilson (2026).

and the main counterfactuals are simulated in the 2011 market environment, the year in which the experimental cohort entered first grade. Online Appendix O-14.1 summarizes how these samples map into each empirical exercise, and Table O-8 reports summary statistics for the main estimation variables.

Our main short-run outcomes are the characteristics of the schools families choose, including distance, fees, test scores, and value added. Our longer-run outcomes are student academic outcomes measured in administrative test-score and attainment records. Because the intervention was delivered during the period in which parents were enrolling children in primary school, timing matters for interpretation. Families reported retrospectively in the follow-up survey that they had already enrolled by the preschool meeting had less scope to respond to the treatment. We therefore report results for both the full sample and the subgroup reporting that enrollment was not yet complete at the meeting, which is also the sample used for the structural treatment-effect estimation. Treatment and control are balanced on most measured characteristics within these subsamples, but not all: the exceptions, including child sex and home ownership, are reported in the balance tables and included in the standard expanded family-control block throughout the reduced-form analysis. Detailed balance tables, additional evidence on enrollment timing, and further discussion of treatment implementation are reported in Table O-5, Table O-6, Table O-7, and Online Appendix Section O-4.

We use the school-level test-score value-added measure developed in Neilson (2026) as our main proxy for academic quality, extending its coverage as needed for the empirical exercises in this paper. This measure captures the school's contribution to later student achievement after conditioning on a rich set of student and school characteristics. We show in the Online Appendix that it is strongly correlated with teacher quality—in particular, with teachers' own scores on Chile's national university entrance exam (PSU, now PAES)—and that rankings are similar whether or not prior scores are available as controls. The report cards given to treated families displayed raw test scores (SIMCE averages), not value added, since raw scores are publicly available and easy for parents to interpret. The two measures are strongly correlated across schools (Figure O-7), so the quality information provided to parents is closely aligned with the measure used in the structural model. The estimation equation and validation exercises are presented in Online Appendix Section O-7.

4. EXPERIMENTAL RESULTS

Table I reports effects on the characteristics of the schools families chose, and Table II reports effects on student outcomes. All specifications include the randomization controls and an expanded family-control block: child sex, SEP priority eligibility, maternal education and marital status, household size, durable-goods ownership, and home ownership, with missing-value indicators. The same block is used for every outcome and does not change the estimation samples. The Online Appendix reports results under alternative control sets and documents control coverage.

Because the intervention was delivered while families were choosing a primary school, we report results for the full sample and by reported enrollment status at the preschool meeting, measured retrospectively in the follow-up survey. Panel B comprises families whose children were already enrolled; Panel C comprises those whose children were not yet enrolled. Enrollment status is observed for 1,571 of the 1,780 children in the analysis sample, so the pooled specification uses a slightly larger sample than Panels B and C combined.

The experimental evidence on school choice is concentrated among families reached before they had committed to a school (Panel C of Table I). In this group, treated families travel about 0.2 km farther, are 12.3 percentage points more likely to choose a fee-charging school, and choose schools with stronger academic profiles.⁸ The clearest shifts are in chosen-school value added and school-average fourth-grade mathematics scores, with a weaker effect on school-average second-grade language scores and an imprecise one on school-average fourth-grade language scores. The value-added effect—about 0.10 in the normalized quality measure used in the model—is the object most directly relevant for the structural analysis. Among families who had already enrolled (Panel B), the estimates are imprecise and none is statistically significant; taken together, they do not show a systematic shift in chosen-school characteristics after the decision had already been made. In the full sample (Panel A), the estimates are attenuated accordingly.

Reported enrollment status refers to a decision made before treatment, but the resulting subsamples need not be identical. Most measured characteristics are balanced within each subgroup, with some exceptions in Panel B, including child sex and home ownership. The expanded

⁸For the distance outcome we restrict to home-to-school distances below 4 km, which excludes families who appear to have moved between the intervention and the following school year. Results are robust to alternative distance thresholds.

TABLE I
EFFECTS ON CHARACTERISTICS OF SCHOOLS CHOSEN ONE YEAR AFTER TREATMENT

	Distance (kms) (1)	Positive Price (2)	Lang 2nd (3)	Lang 4th (4)	Math 4th (5)	V. Added ^a (6)
Panel A: Full Sample						
Treatment	0.063 (0.062)	0.050 (0.034)	0.009 (0.020)	0.013 (0.024)	0.021 (0.026)	0.035 (0.030)
Control mean	1.081	0.494	-0.119	-0.095	-0.082	-0.024
N obs.	1,427	1,701	1,693	1,692	1,692	1,701
Panel B: Already enrolled at the time of the PreK visit						
Treatment	-0.161 (0.133)	0.000 (0.055)	-0.026 (0.041)	-0.021 (0.047)	-0.042 (0.056)	-0.025 (0.058)
Control mean	1.192	0.589	-0.031	-0.007	0.046	0.067
N obs.	512	578	576	577	577	578
Panel C: Not enrolled at the time of the PreK visit						
Treatment	0.198*** (0.073)	0.123*** (0.040)	0.052* (0.028)	0.043 (0.029)	0.082** (0.033)	0.099*** (0.037)
Control mean	1.010	0.434	-0.179	-0.137	-0.146	-0.067
N obs.	800	934	928	926	926	934

Notes: ITT estimates from regressions of chosen-school characteristics on treatment assignment. Panel A: full sample; Panel B: families already enrolled at the time of treatment; Panel C: families not yet enrolled. All specifications include randomization controls and the expanded family controls listed in Appendix A1. Standard errors in parentheses are clustered at the preschool-center level. ^a Column 6 uses the 2011 school value-added measure Q_2 from Neilson (2026). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

family controls account for these observed differences and improve precision. Treatment does not predict enrollment-status classification or its observability.⁹ The not-yet-enrolled group is also the policy-relevant margin under Chile's current centralized admissions system, in which families face a common application window and oversubscribed seats are allocated using cen-

⁹Older siblings may constrain school choice. Linked administrative records show an older sibling in grades 1–8 for 34.6% of already-enrolled and 33.3% of not-yet-enrolled families. Conditional on the controls, sibling status predicts a –0.9 percentage-point difference in prior enrollment (s.e. 2.6), and adding sibling status leaves the estimated school-choice effects essentially unchanged. The Online Appendix reports the full comparison.

trally specified priorities and lottery numbers; the structural analysis therefore focuses on this group.

Table II examines whether the changes in school choice translated into better academic outcomes. Among not-yet-enrolled families, fourth-grade SIMCE scores are 0.24 SD higher in language and 0.21 SD higher in mathematics. The eighth-grade estimates are 0.18 and 0.26 SD, respectively. The treatment does not affect reaching twelfth grade or participation in the college entrance exam. Conditional on taking the PAES, the language and mathematics estimates are 0.19 and 0.15 SD.

The later outcomes are observed for smaller and potentially selected test-taking samples. Students remain observable in administrative enrollment records, but not all students sit each assessment, and the 2020 eighth-grade SIMCE was cancelled for one cohort. Treatment does not predict score observability, either overall or within the enrollment-status groups. Nevertheless, because test participation selects the observed samples and substantially reduces their size, we interpret the eighth-grade and PAES estimates with appropriate caution.

This qualification is particularly relevant to Panel B. The long-horizon estimates for already-enrolled families are imprecise. Under the expanded-control specification, neither PAES coefficient is statistically significant. Eighth-grade mathematics remains positive and significant, but it is estimated on 243 children drawn almost entirely from one cohort. It is also the only one of the 14 school-choice and student-outcome coefficients in Panel B that rejects zero at the 5-percent level. For reference, if all 14 null hypotheses were true and the tests were independent, the probability of at least one such rejection would be $1 - 0.95^{14} = 0.51$. More formally, the Holm-adjusted p -value for the eighth-grade mathematics estimate across the 14 outcomes is 0.319. The Online Appendix reports alternative control sets and detailed coverage. Taken together, these estimates do not establish a distinct long-run mechanism for already-enrolled families. The structural analysis uses the treatment-induced school-choice response among not-yet-enrolled families; achievement coefficients do not enter its estimation or counterfactuals.

Finally, we examine choices in relation to the report card's design features. As reported in the Online Appendix, treated families are less likely to choose a listed school. Conditional on choosing a listed school, the effect on choosing a green-coded school is small and statistically insignificant, while the high-ranked-school indicator increases among families who had not yet enrolled. This conditional association does not separately identify a causal effect of highlighting: school rank and visual presentation were not independently randomized, and choosing a listed school is itself affected by treatment.

TABLE II
EFFECTS ON STUDENT OUTCOMES

	SIMCE Test Scores				Reached 12th Grade (5)	College Entrance Exam (PAES)		
	4th Grade		8th Grade			Took PAES (6)	(Cond. on taking PAES)	
	Lang (1)	Math (2)	Lang (3)	Math (4)			Lang (7)	Math (8)
<i>Panel A: Full Sample</i>								
Treatment	0.096* (0.052)	0.109** (0.054)	0.034 (0.082)	0.235** (0.094)	-0.004 (0.029)	0.027 (0.030)	0.166** (0.072)	0.133** (0.053)
Control mean	-0.222	-0.145	-0.059	-0.103	0.728	0.592	-0.367	-0.225
N obs.	1,583	1,584	610	615	1,775	1,775	1,083	1,060
<i>Panel B: Already enrolled at the time of the PreK visit</i>								
Treatment	-0.082 (0.108)	-0.119 (0.098)	-0.195 (0.229)	0.450** (0.193)	-0.036 (0.049)	0.013 (0.054)	0.165 (0.109)	0.159 (0.098)
Control mean	-0.039	0.058	0.169	0.044	0.770	0.613	-0.287	-0.145
N obs.	543	533	241	243	596	596	373	364
<i>Panel C: Not enrolled at the time of the PreK visit</i>								
Treatment	0.241*** (0.071)	0.209*** (0.068)	0.184 (0.112)	0.256** (0.115)	0.002 (0.036)	0.029 (0.031)	0.187** (0.082)	0.147** (0.071)
Control mean	-0.339	-0.252	-0.188	-0.162	0.706	0.579	-0.447	-0.284
N obs.	857	865	301	301	975	975	581	568

Notes: ITT estimates from regressions of student outcomes on treatment assignment. Panel A: full sample; Panel B: families already enrolled at the time of treatment; Panel C: families not yet enrolled. Columns 1–2: 4th-grade SIMCE; Columns 3–4: 8th-grade SIMCE; Columns 5–6: reaching twelfth grade and college entrance exam participation; Columns 7–8: PAES scores, conditional on taking the exam. The eighth-grade samples consist almost entirely of the 2012 entry cohort because the 2020 national assessment was cancelled. Treatment does not predict eighth-grade mathematics-score observability in Panel B ($p = 0.838$). Eighth-grade mathematics is the only one of the 14 Panel-B school-choice and student-outcome coefficients that rejects zero at the 5-percent level; its Holm-adjusted p -value over the 14 outcomes is 0.319. All specifications include randomization controls and the expanded family controls listed in Appendix A1. Standard errors in parentheses are clustered at the preschool-center level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In sum, personalized information shifts families toward higher-quality schools and raises student achievement among families reached before enrollment. Positive estimates also appear in later observed assessment samples, although those samples are smaller and selected by test participation. The treatment-induced shifts in chosen school quality, price, and distance provide the experimental targets for the structural analysis in the next section.

5. EMPIRICAL MODEL FOR POLICY ANALYSIS

This section develops the structural model. We adopt the baseline demand system—utility specification, market shares, and family types—from Neilson (2026) and summarize it briefly in subsection 5.1. Our main contribution on the demand side is a treatment-effects parameterization (subsection 5.2) that allows the information intervention to shift school choice through characteristic-specific channels and a common rescaling of choice behavior. A Bayesian signal model (subsection 5.3) provides one interpretation of the scale parameter, following Brown and Jeon (2024). On the supply side (subsection 5.4), we recover cost parameters from schools’ quality first-order conditions and introduce a capacity extension that modifies the quality response when schools cannot easily expand enrollment. subsection 5.5 describes the three-step estimation.

5.1. *Baseline Demand Model*

Following Neilson (2026), the utility for family i from school j is

$$U_{i,j} = \bar{\beta}x_j + \xi_j + \beta_i q_j + \alpha_i \text{op}_{k(i),j} + \lambda_i d_{\text{loc}(i),j} + \epsilon_{i,j}, \quad (1)$$

where q_j is school academic quality (value added), $\text{op}_{k(i),j}$ is the out-of-pocket payment for a family of type k at school j given the prevailing voucher policy, and $d_{\text{loc}(i),j}$ is distance from the family’s residential location to school j . The vector x_j collects observable school characteristics over which families share common preferences $\bar{\beta}$: administration type, for-profit status, religious affiliation, and whether the school serves grades K through 12. The term ξ_j is the standard BLP-style unobserved school attribute, and $\epsilon_{i,j}$ is i.i.d. extreme value type 1. The distinction between the school’s posted tuition p_j and the family’s out-of-pocket payment $\text{op}_{k(i),j}$ matters for the supply side: schools set a single tuition, while out-of-pocket payments depend

on how that tuition interacts with the voucher schedule; in particular, SEP-eligible families at SEP-participating schools pay zero out of pocket.¹⁰

Preferences are heterogeneous across six discrete family types $k \in \{1, \dots, 6\}$, defined by mother's education and SEP eligibility (Neilson, 2026). Quality preferences include a type-specific component β_k and a standard-normal draw $z_i \sim N(0, 1)$, so that $\beta_i = \beta_{k(i)} + \sigma z_i$, while price and distance coefficients are type-specific: $\alpha_i = \alpha_{k(i)}$ and $\lambda_i = \lambda_{k(i)}$. We use signed price coefficients: a negative α_k means that a higher out-of-pocket payment lowers utility. Types 1 and 3 are SEP-eligible, types 2 and 4 are not, and types 5–6 are the two higher-education types.

Families choose the school that maximizes $U_{i,j}$ from the set of schools in their market m . The market share of school j aggregates over types, residential locations, and the random coefficient:

$$s_j(\mathbf{q}, \mathbf{op}, \theta) = \sum_{k=1}^K \Pi_k^m \sum_{\text{loc}}^{L^m} w_k^{\text{loc}} \int s_j^{\text{loc},k}(\theta, z) d\Phi(z), \quad (2)$$

where Π_k^m is the proportion of type k families in market m , w_k^{loc} is the within-type distribution across locations, and $s_j^{\text{loc},k}(\theta, z)$ is the standard logit choice probability conditional on type, location, and standard-normal draw, with Φ denoting the standard-normal distribution. See Neilson (2026) for the full derivation, estimation, and resulting parameter estimates $\theta = \{\bar{\beta}, \alpha, \beta, \lambda, \sigma, \xi\}$.

5.2. Treatment Effects Parameterization

The information treatment may affect school choice through multiple channels. Providing families with school-specific report cards can shift the weight they place on school quality, price, and distance. It can also sharpen choice behavior more broadly, increasing families' responsiveness to all school attributes. We parameterize both channels and let the data determine their relative importance.

Define the baseline (control) deterministic utility as

$$V_{ij}^C = \tilde{\delta}_j + \bar{\phi}_k^{q,C} q_j + \bar{\phi}_k^{op,C} \text{op}_{ij} + \bar{\phi}_k^{d,C} d_{ij} + \bar{\sigma}^C z_i q_j, \quad (3)$$

where $\delta_j \equiv \bar{\beta}^q q_j + \bar{\beta}^C x_j + \bar{\xi}_j^C$ denotes the full recovered school mean utility from Neilson (2026), and $\tilde{\delta}_j \equiv \delta_j - \bar{\beta}^q q_j$ removes its common quality component. Here $\bar{\phi}_k^{q,C}$ is the full type-

¹⁰We use “location” for readability; these are the same discrete residential nodes used in Neilson (2026).

specific quality coefficient, including $\bar{\beta}^q$, while $\bar{\phi}_k^{op,C}$ and $\bar{\phi}_k^{d,C}$ are the signed price and distance coefficients. The draw z_i is standard normal and $\bar{\sigma}^C$ is its baseline quality-taste dispersion.¹¹ The treated utility adds two components:

$$V_{ij}^T = \kappa_k \cdot V_{ij}^C + \psi_k^q q_j + \psi_k^{op} op_{ij} + \psi_k^d d_{ij}. \quad (4)$$

The first term rescales all baseline utility components by a common factor κ_k —including residual mean utility $\tilde{\delta}_j$, which contains the common preferences $\bar{\beta}^C x_j$ and the unobserved school attribute $\bar{\xi}_j^C$. The ψ_k^c terms add characteristic-specific treatment effects on top.

The model restricts treatment effects on the coefficients of other school attributes, including sector and residual school attributes, to the common rescaling:

$$\bar{\beta}_k^T = \kappa_k \cdot \bar{\beta}_k^C. \quad (5)$$

For attributes targeted by the information intervention—quality, out-of-pocket price, distance—the effective coefficient shifts through both the common scale and a characteristic-specific treatment effect:

$$\bar{\phi}_k^{c,T} = \kappa_k \cdot \bar{\phi}_k^{c,C} + \psi_k^c. \quad (6)$$

Equation 5 rules out separate treatment shifts in these coefficients. Voucher-school attendance and residualized mean-utility moments therefore add information about κ_k and the characteristic-specific shifts. Both channels change choice probabilities and can change these chosen-school outcomes. In practice, κ_k is not pinned down by any single moment but is jointly disciplined by the full system of moments. When $\kappa_k = 1$, the model reduces to characteristic-specific treatment effects with no common scaling, nesting the standard (unscaled) specification.

Interpretation. The treatment-effect parameters ψ_k^c are summary parameters that capture characteristic-specific shifts in how much characteristic c influences school choice, beyond common rescaling. Each ψ_k^c combines two mechanisms that are observationally equivalent in our setting. The first is an *information channel*: more precise signals raise the weight families place on the corresponding characteristic in expectation, even if underlying preferences are un-

¹¹Under the standard logit normalization, these estimates represent structural preference parameters divided by the error scale. subsection 5.3 develops this point formally; the normalization is innocuous for the baseline demand but becomes consequential when treatment may change the scale.

changed (Brown and Jeon, 2024). The second is a *preference channel*: by making quality differences salient and easy to compare, the report cards may cause families to place more weight on academic quality independent of any change in signal precision. The experiment identifies the combined effect of these channels on school choice; separately identifying them would require data on subjective beliefs.

For the policy counterfactuals in this paper, separately identifying these underlying channels is not necessary. The counterfactual exercises apply the same treatment observed in the experiment to eligible families (types 1–4) at full take-up, and simulate the resulting equilibrium. The demand shifts we estimate are identified directly from experimental variation, so the estimated total shift—however it decomposes across the information and preference channels—is the relevant sufficient statistic for these simulations.

To allow for heterogeneous treatment effects, we estimate separate ψ_k^c parameters and scale parameters κ_k for SEP-eligible families ($k \in \{1, 3\}$) and non-SEP families ($k \in \{2, 4\}$). The full treatment parameter vector is

$$\theta_2^T = \left(\psi_{\text{SEP}}^q, \psi_{\text{SEP}}^{op}, \psi_{\text{SEP}}^d, \psi_{\text{nSEP}}^q, \psi_{\text{nSEP}}^{op}, \psi_{\text{nSEP}}^d, \kappa_{\text{SEP}}, \kappa_{\text{nSEP}} \right). \quad (7)$$

We incorporate the treatment into the demand model by creating *treated types* in addition to the baseline six types from Neilson (2026). The experimental sample includes families of types 1–4 (types 5–6 have too few observations in the experiment). Treated families have utility V_{ij}^T from Equation 4, while control families retain the baseline utility V_{ij}^C .

5.3. Imperfect Information and the Scaled Logit

The parameterization in subsection 5.2 can be interpreted through a Bayesian signal model of school characteristics. When families observe school quality, price, and distance with noise, the posterior mean for each characteristic is attenuated toward the prior by a signal-to-noise ratio $\rho_k^c \in (0, 1)$ that depends on signal precision. The effective preference weight on characteristic c becomes a product of the structural preference and the signal precision: $\phi_k^c = \beta_k^c \cdot \rho_k^c$. Crucially, as emphasized by Brown and Jeon (2024), signal noise also enters the composite error term of the choice model. Improving signal precision on specific characteristics can therefore change not only the weight on those characteristics but also the error variance of the entire choice

model. We adopt a scaled logit approximation that captures this possibility while preserving the logit functional form.¹²

Dividing the expected utility by the type-specific error scale μ_k (derived in the appendix from the composite error variance) recovers standard logit choice probabilities with *effective* parameters that embed both preferences and signal precision. The baseline demand estimation in Neilson (2026) identifies only these effective parameters under the standard normalization $\mu_k^C = 1$. This normalization is innocuous for the baseline demand but becomes consequential when treatment may change the scale. Within this framework, the common scale parameter from subsection 5.2 is

$$\kappa_k \equiv \frac{\mu_k^C}{\mu_k^T}. \quad (8)$$

When improved signal precision reduces the composite error variance, $\mu_k^T < \mu_k^C$ and $\kappa_k > 1$: the composite error shrinks, sharpening the weight on all school attributes, including those about which the treatment provides no new information. The model nests the standard logit at $\kappa_k = 1$, i.e., when treatment does not change the error variance. We do not interpret $\kappa_k > 1$ as parents literally learning about every residual school characteristic; rather, it is a parsimonious way to allow for any common sharpening of choice behavior beyond the characteristic-specific treatment shifts. Appendix A3 provides a simplified type-level illustration, including the posterior, expected utility, composite error, and variance-matching formulas.

5.4. Supply Side

We model privately owned schools as choosing quality q_j to maximize a weighted objective that reflects both profit motives and other institutional goals. On the supply side, the relevant price variable is the school's posted tuition p_j (the sticker price). Family out-of-pocket payments op_{ij} are derived from that posted tuition after applying the voucher schedule and therefore vary by student type and school participation status. Revenue per student depends on the sticker price and the voucher schedule $v_j^k(p_j)$, where j indexes schools and k indexes student type. In the baseline demand system, posted tuition is treated as an endogenous observed school characteristic: schools that invest in unobserved quality attributes may also set higher prices, inducing

¹²This variance-matching approximation is standard in the heteroskedastic logit literature; see (Train, 2009), Chapters 2 and 6. It is most accurate when the EV1 taste shock dominates the signal noise, a condition supported by our empirical finding that the scale correction is modest (κ not significantly different from 1).

a correlation between tuition and the unobserved school attribute ξ_j . The demand estimation addresses this through instrumental variables, so the recovered ξ_j already accounts for this correlation. For the supply-side exercise, we take tuition and the estimated ξ_j as given and do not model a pricing first-order condition. The reason is institutional: during the period used to estimate the supply model, subsidized private schools operated in a regulated and segmented pricing environment rather than under standard unconstrained price setting.¹³ The Online Appendix displays the statutory schedule and the distinction between posted tuition and families' out-of-pocket payments. Because our objective here is to study the medium-run quality response to the information policy, and given the institutional restrictions on copayments described in section 3, we hold observed tuition fixed in the supply exercise and focus on the quality first-order condition as the relevant adjustment margin. This does not imply that historical tuition was exogenous; it defines the adjustment margins included in the counterfactual. The per-student cost of providing quality q_j is

$$C_{jt}(q_{jt}) = \gamma_{\text{SEP}} \text{SEP}_{jt} + \gamma_{mt}^{\text{type}(j)} + c_{jt} q_{jt}, \quad c_{jt} = c_j + \delta_{\text{type}(j)}(t - 2006). \quad (9)$$

Here SEP_{jt} indicates program participation, $\gamma_{mt}^{\text{type}(j)}$ is a market-year effect specific to voucher or private non-voucher schools, c_j is a school-specific unit cost of quality, and $\delta_{\text{type}(j)}$ allows that unit cost to trend differently by school type.¹⁴ The first-order condition for optimal quality at an unconstrained school yields

$$q_{jt}^* = \underbrace{\left[\frac{R_{jt} - \gamma_{\text{SEP}} \text{SEP}_{jt} - \gamma_{mt}^{\text{type}(j)}}{c_{jt}} \right]}_{\text{Competitive quality}} - \underbrace{s_{jt} \left[\frac{\partial s_{jt}}{\partial q_{jt}} \right]^{-1}}_{\text{Quality markdown}}, \quad (10)$$

where R_{jt} is observed revenue per student (the regular subsidy, SEP subsidy, and posted tuition, as applicable). This expression decomposes optimal quality into a competitive component and a markdown reflecting market power. The markdown $s_{jt}/(\partial s_{jt}/\partial q_{jt})$ is inversely related to the semi-elasticity of demand with respect to quality: schools with more elastic demand face

¹³Under the general shared-financing regime in DFL No. 2 of 1998, schools could charge positive copayments but faced caps and a reduction in the general subsidy as average fees rose. SEP then added a further layer of regulation by requiring participating schools to charge zero copayments to priority students (Law No. 20.248, Art. 6). See also Sanchez (2017) for related discussion of strategic school responses under Chile's voucher design.

¹⁴The quality index q_j recovered from the demand estimation is identified only up to differences. In the supply estimation, we normalize q_j so that its minimum is zero and impose a small positive floor on c_{jt} .

stronger competitive pressure and provide quality closer to the competitive level. Neilson (2026) derives this quality FOC and uses it to interpret the observed response to the SEP reform. Because the information intervention has not been implemented at scale, we instead recover the cost parameters required to predict counterfactual quality choices. We also extend the supply problem to incorporate the capacity and revenue-composition channel described below.

5.4.1. *Interpreting Supply-Side Parameters*

Schools in the data belong to three administrative sectors: municipally managed public schools, privately managed voucher schools, and private non-voucher schools. Within the voucher sector, an administrative indicator records whether the operator's legal form was for-profit or nonprofit; the latter include foundations and religious organizations, including Catholic providers. This was a substantive legal distinction during our estimation period; the 2015 Inclusion Law subsequently required subsidized private providers to become nonprofit. Legal form need not, however, reveal the provider's complete objective. Following Hoxby (2003), we distinguish the distribution of operating surplus from the incentive to generate resources that can be used by the school.

To make this distinction precise, let operating surplus be

$$\Pi_j(q_j) = n_j(q_j)[R_j - C_j(q_j)] - F_j.$$

A for-profit provider may maximize Π_j . If a nonprofit provider instead values both operating surplus and educational quality delivered to its students,

$$V_j(q_j) = a_j \Pi_j(q_j) + (1 - a_j) n_j(q_j) B_j(q_j), \quad a_j > 0,$$

then its problem can be written in the same form as surplus maximization after replacing input cost with the effective cost

$$\tilde{C}_j(q_j) = C_j(q_j) - \frac{1 - a_j}{a_j} B_j(q_j).$$

When the mission payoff is per student, linear in quality, and unchanged by the information policy, direct concern for quality changes the net marginal cost of providing it but leaves the form of the quality FOC in Equation 10 unchanged. The school-specific term c_j can therefore reflect persistent differences in both input costs and schools' objective functions. Type-specific

trends allow this term to evolve differently for voucher and private non-voucher schools, and the supply instruments are interacted with the administrative for-profit and nonprofit-voucher indicators.

For the counterfactuals, we write the quality FOC in changes around the observed equilibrium. Equivalently, we use observed quality to recover the school-specific unobservable that rationalizes each school's choice and hold it fixed when the policy shifts demand. Capacity determines which margin governs the response: unconstrained schools can attract additional enrollment, whereas schools near capacity can change the composition of enrollment. Under either regime, changes in quality are driven by the policy-induced change in the marginal return to quality, holding fixed persistent differences in input costs and objective functions.

This argument does not cover every possible nonprofit objective. A mission payoff unrelated to the number of students served, depending directly on student composition, determined by a binding break-even constraint, or changing with the information policy generally adds terms to the FOC that need not remain fixed when demand changes. We therefore do not interpret c_j as a deep technological parameter. Separately identifying schools' production technology and objective functions is beyond the scope of this paper.

Public schools are different institutionally. Municipalities manage portfolios of schools, can pool resources and supplement per-pupil transfers, and face rigidities in adjusting teaching staff. The school-level budget constraint is therefore softer than for private providers (Gallego, 2013). We exclude public schools from supply estimation and hold their quality fixed in the main counterfactual, treating this as a scope restriction rather than assuming that a private-provider FOC describes municipal decisions.

5.4.2. *Capacity Constraints and the Supply-Side FOC*

The baseline FOC in Equation 10 corresponds to the standard case in which a school can expand enrollment when higher quality raises demand. This is the same approximation used in the baseline demand system estimated by Neilson (2026), who argues that capacity constraints were not a first-order feature of household choice in the historical estimation period.¹⁵ We maintain that approximation for the baseline demand inputs, but allow for the possibility that a

¹⁵Neilson (2026) argues in the Online Appendix that this approximation is reasonable in the baseline data because primary-level selection by voucher schools was illegal during the estimation period, only 2% of urban primary schools reached the legal class-size cap, and only 2% of surveyed parents reported preferring another school but being turned away.

subset of schools nonetheless faced limited room to expand enrollment on the supply side, and that such constraints become more relevant in the at-scale policy counterfactuals studied below.

Capacity-constrained schools may still have incentives to improve quality through a *composition channel*. Under Chile’s Deferred Acceptance assignment mechanism, SEP-eligible students receive priority at schools that participate in the SEP program. When a school near capacity raises its quality, it need not expand total enrollment to benefit: it can instead attract a more SEP-intensive student body. Since SEP students generate higher per-student revenue ($v_b + v_{\text{SEP}}$ versus $v_b + p_j$), this compositional shift creates a financial incentive for quality improvement even when the standard enrollment channel is muted.

Formally, for a school operating at capacity $\bar{c}_j$, the first-order condition for quality becomes

$$c_{jt} = \Delta R_{jt} \cdot \frac{\partial s_{jt}^{\text{SEP}} / \partial q_{jt}}{s_{jt}}, \quad (11)$$

where $\Delta R_{jt} = R_{jt}^{\text{SEP}} - R_{jt}^{\text{nonSEP}}$ is the per-student revenue gap between SEP and non-SEP students, and $\partial s_{jt}^{\text{SEP}} / \partial q_{jt}$ is the derivative of the SEP-weighted share with respect to quality. The right-hand side captures the marginal revenue gain from improving composition: higher quality attracts more SEP students (who generate ΔR_{jt} more revenue each), scaled by the fraction of the student body affected. Under the linear cost function, the marginal cost of quality is c_{jt} , so the FOC equates this marginal cost to the marginal composition benefit.

In practice, we classify each school-year observation as operating under one of the two regimes—unconstrained or capacity-constrained—based on observed enrollment relative to capacity, and estimate the supply parameters using both FOCs simultaneously with the school-specific cost parameter c_j shared across regimes. Section 5.5.5 describes the classification rule and estimation details.

5.5. Estimation

Our estimation proceeds in three steps: (i) recovering baseline demand parameters from Neilson (2026), (ii) estimating treatment effect parameters using the experimental variation, and (iii) recovering supply-side cost parameters.¹⁶ We begin with an informal discussion of identification.

¹⁶See Appendix A4 for additional estimation details and standard error formulas.

5.5.1. Identification

Baseline demand. The baseline demand parameters $\theta = \{\bar{\beta}, \alpha, \beta, \lambda, \sigma, \xi\}$ are identified by three complementary sources of variation in Neilson (2026): the BLP inversion (Berry et al., 1995) recovers full school-level mean utilities δ_j by matching predicted to observed market shares; micro-moments matching the average quality, out-of-pocket payment, and distance chosen by each family type identify heterogeneity across the six family types (Petrin, 2002); and IV moments address the endogeneity of price and quality using policy-driven revenue variation from the voucher and SEP reforms and local labor-cost shifters that proxy for the cost of hiring skilled workers. Appendix A4 lists these moments and instruments explicitly. As discussed in subsection 5.3, the baseline estimates identify the *effective* parameters that embed the normalization $\mu_k^C = 1$.

Treatment effects on informed characteristics. The ψ_k^c parameters — capturing how much treatment shifts the weight on quality, price, and distance beyond the common scale effect — are identified by the random assignment of the information treatment. Preschool centers were randomly assigned to treatment or control, stratified by region, grades offered, and the number of primary schools within 2 km. Treatment effects compare families across those assigned centers; preschool-municipality fixed effects and market controls are regression adjustments. The moments match these experimental treatment effects, separately for each characteristic and type group.

Scale parameter. The scale parameters κ_g and characteristic-specific shifts are jointly disciplined by the full system of 12 moment entries. Voucher-school attendance and residualized mean utility $\tilde{\delta}_j$ provide information about choices along dimensions whose coefficients have no separate treatment shift. Changes in quality, price, or distance weights also change the probabilities of choosing schools with these attributes. Together with the treatment effects on informed characteristics (q , op , d), these moments discipline the 8-parameter vector; no single moment identifies κ_g in isolation.

Supply. Supply parameters are identified from the first-order conditions for quality (Equation 10 and Equation 11). SEP participation is the observable cost shifter in Equation 9; it is distinct from the instruments used to identify its coefficient. Those instruments carry over the excluded variation used in the baseline demand estimation—simulated transfer changes, reform timing interacted with pre-policy neighborhood SEP exposure, and local non-education earn-

ings deviations—and add an inverse-distance-weighted index of nearby public-school teacher quality. The implementation also interacts the instrument matrix with for-profit and nonprofit-voucher indicators. School-specific quality costs and market-year-type effects are concentrated out, leaving γ_{SEP} and the voucher- and private-school cost trends as the three freely estimated parameters.

5.5.2. *Step 1: Baseline Demand Parameters*

The baseline demand parameters are taken from Neilson (2026), who estimates the first-step demand system using administrative first-grade choice data from 2006, 2007, 2011, and 2012. We refer the reader to that paper for a detailed description of the estimation strategy and the resulting parameter estimates.

5.5.3. *Estimation Sample*

The treatment effect parameters are estimated using a subsample of the experimental families that can be matched to the market-level demand model. We begin with the full experimental sample and apply a series of restrictions, each motivated by the requirements of the structural estimation. First, we restrict attention to families that had *not yet enrolled* their children at the time of the Pre-K visit, since the treatment effects on school choice are concentrated entirely among these families (section 4). Second, we restrict to the six urban markets that overlap between our experiment and the market structure in Neilson (2026), which is necessary because the baseline demand parameters—mean utilities δ_j , quality measures, and spatial choice sets—are only available within Neilson’s market definitions. Third, we apply a geographic consistency filter: we keep families if they remain in the same municipality as at baseline or if the chosen school is within 4 km of the family’s home, excluding only observations that are both in a different municipality and implausibly far from the chosen school.

We then drop families with missing information on the school they ultimately chose (value-added, out-of-pocket price, or distance). Finally, family types are assigned using six categories defined by mother’s education and SEP eligibility; because the experiment contains few families of types 5–6 (technical and college-educated), we collapse these into types 3–4 using actual JUNAEB SEP eligibility records to ensure correct classification.

Table III reports the treatment effects on chosen school characteristics in the resulting estimation sample, separately for SEP-eligible and non-SEP families. These are the experimental moments that the structural model must match. The treatment significantly increases the

value-added of schools chosen by SEP families (0.220, $p < 0.05$); the point estimate on driving distance is also positive (0.134 km) though not statistically significant. These treatment effects form the empirical targets for the GMM estimation described below.

TABLE III
TREATMENT EFFECTS IN THE MODEL SAMPLE (EXPERIMENT MOMENTS)

	Quality	Sticker Price	OOP Price	Distance	Voucher	N
Pooled	0.195*** (0.073)	0.039** (0.017)	0.027 (0.017)	0.118 (0.078)	0.114*** (0.042)	815
SEP	0.220** (0.087)	0.025 (0.017)	0.006 (0.016)	0.134 (0.092)	0.106** (0.043)	505
Non-SEP	0.114 (0.104)	0.048 (0.036)	0.048 (0.036)	0.184 (0.119)	0.096 (0.067)	310
Preschool-municipality FE	Yes	Yes	Yes	Yes	Yes	

Notes: Experimental moments used in the demand estimation. Regressions use the model-overlap sample described in Section 5.5.3, preschool-municipality fixed effects, and standard errors clustered at the preschool-center level; exact controls are summarized in Appendix A2. Quality is normalized value added Q_2 . Distance is Google Maps driving distance, restricted to observations within 4 km. Voucher indicates a private voucher rather than a public school. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.5.4. Step 2: Treatment Effect Parameters

Conditional on the Step 1 estimates, we estimate the treatment parameter vector θ_2^T defined in Equation 7 using GMM. The identification strategy exploits the random assignment of the information treatment within the experimental sample.

We define *RCT moments* that match the treatment effects on school characteristics observed in the experimental data. For each characteristic $c \in \{q, p^{\text{sticker}}, \text{op}, d\}$ and each type group $g \in \{\text{SEP}, \text{nonSEP}\}$, the moment condition requires the model-simulated treatment effect to match the experimental estimate:

$$G_{c,g}(\theta_2^T) = \hat{\beta}_{c,g}^{\text{RCT}} - \hat{\beta}_{c,g}^{\text{sim}}(\theta_2^T), \quad (12)$$

where $\hat{\beta}_{c,g}^{\text{RCT}}$ is the coefficient on the treatment indicator from a regression of characteristic c on treatment status (with preschool-municipality fixed effects and the controls listed in Appendix A2) in the experimental data, and $\hat{\beta}_{c,g}^{\text{sim}}(\theta_2^T)$ is the analogous coefficient computed from model-simulated school choices. The simulated choices are generated using the treated and control probabilities implied by the model, with the random coefficient integrated via Gauss-

Hermite quadrature. To help discipline the scale parameters κ_g , we add moments on voucher-school attendance and residualized mean utilities $\tilde{\delta}_j \equiv \delta_j - \bar{\beta}^q q_j$, both of which capture treatment effects on already-known school attributes. Their coefficients have no separate treatment shift under Equation 5, but their chosen averages respond to both common scaling and characteristic-specific shifts through choice probabilities. Together, the full system of moments jointly disciplines all eight parameters (seven freely estimated in the benchmark specification, with $\psi_{\text{SEP}}^{\text{op}}$ restricted to zero as discussed in the results section).

We estimate using two-step efficient GMM: the first step uses the identity weighting matrix, and the second step uses $W = \hat{V}_{\text{CR}}^{-1}$, where $\hat{V}_{\text{CR}}$ is the cluster-robust covariance matrix of the data moments (clustering at the preschool center level). Standard errors are computed using the standard GMM sandwich formula.

5.5.5. Step 3: Supply Parameters

Given the demand parameters from Steps 1 and 2, we estimate the supply-side cost parameters using GMM on the quality first-order conditions. We classify a school-year observation as capacity-constrained when its enrollment-to-capacity ratio exceeds 0.95 and the school receives differential SEP revenue. This classification uses observed enrollment and imputed capacity and should therefore be understood as a transparent empirical rule rather than an exact measure of when capacity begins to bind. In our preferred specification, 2,414 of 22,882 school-year observations (10.5%) are classified as constrained, so the two-regime extension modifies the supply problem for a meaningful but limited subset of the sample.

For each school-year observation, we compute the quality markdown $\text{MD}_{jt} = s_{jt}/(\partial s_{jt}/\partial q_j)$ from the estimated demand system and define the “competitive quality” residual:

$$\text{CompQ}_{jt} \equiv q_{jt} + \text{MD}_{jt} = \frac{R_{jt} - \gamma_{\text{SEP}} \text{SEP}_{jt} - \gamma_{mt}^{\text{type}(j)}}{c_{jt}}, \quad (13)$$

where $c_{jt} = c_j + \delta_{\text{type}(j)}(t - 2006)$ and $\gamma_{mt}^{\text{type}(j)}$ absorbs common shocks within each market, year, and school type. This is the left-hand-side object implied by rearranging the unconstrained quality FOC; it is not a direct measure of “quality investment” or mean utility. For capacity-constrained observations, the estimation instead uses the composition-based condition in Equation 11.

We concentrate out c_j and $\gamma_{mt}^{\text{type}(j)}$ jointly. The three freely estimated parameters are γ_{SEP} , which captures the additive cost differential associated with SEP participation, and the voucher-

and private-school trends in the unit cost of quality. The estimation sample consists of voucher and private non-voucher schools observed over 10 years (2006–2015) across 53 markets (22,882 school-year observations, 2,389 schools). Public schools are excluded for the reasons discussed in Section 5.4.1.

The moment conditions interact the regime-specific structural residual Ω_{jt} with the instrument matrix. This matrix carries over the baseline demand instruments and predetermined school characteristics, adds interactions with for-profit and nonprofit-voucher indicators, and includes a new Nearby Teacher Quality Index—the inverse-distance-weighted average of public-school teacher PSU scores within each market-year. Table O-15 documents the relationship between these instruments and observed school quality; the table is an instrument-relevance diagnostic rather than the supply GMM equation itself. We estimate using two-step optimal GMM. We view this two-regime supply specification as a disciplined extension of the baseline model rather than a fully unified model of demand, assignment, and capacity: it uses the same latent-demand primitives as the rest of the paper, but lets the marginal incentive to raise quality depend on whether a school appears to have slack enrollment capacity.

5.6. *Parameter Estimates*

Table IV presents the demand-side parameter estimates. Panel A reports the baseline demand parameters from Neilson (2026); Panel B reports the treatment effect parameters estimated in this paper. The baseline model is defined over six family types based on mother’s education and SEP eligibility. Our counterfactual applies the intervention only to types 1–4—families in which the mother has at most completed high school, separately by SEP eligibility—and leaves the two higher-education types untreated. All four treated types are substantially represented in the 815-family structural experiment sample: types 1–4 contain 265, 92, 240, and 218 families, respectively, with both treatment and control observations in every type. Because treatment heterogeneity is summarized more parsimoniously by SEP status, the experimental demand shift is identified from 505 SEP families (types 1 and 3) and 310 non-SEP families (types 2 and 4).

The experimental type shares need not equal their population shares. The counterfactual combines the experimentally estimated demand shift with the target population’s own distribution of family types, residential locations, and available schools. Administrative data for an adjacent Integra cohort provide a useful support check: 90.1 percent of Integra children entering first grade in the 53 model markets belong to types 1–4, and each of the four types is well

TABLE IV
DEMAND MODEL ESTIMATES

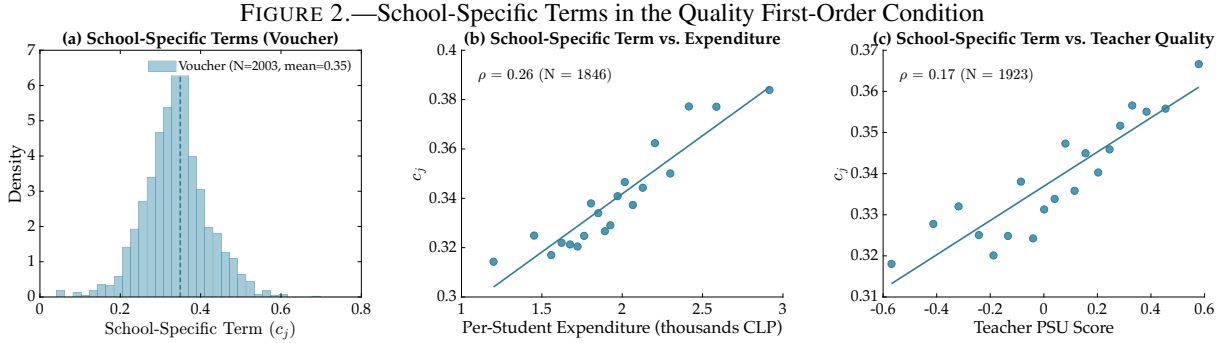
	<i>Panel A: Baseline Demand (Neilson 2026)</i>				<i>Panel B: Treatment Effects (this paper)</i>		
	Quality (β^q)	Price (α)	Distance (λ)		Quality (ψ^q)	Price (ψ^{op})	Distance (ψ^d)
<HS	1.40 (0.03)	−2.70 (1.10)	−1.29 (0.05)	SEP	0.500 (0.230)	—	0.427 (0.204)
HS	0.56 (0.10)	−0.56 (0.38)	−1.09 (0.04)	nonSEP	0.399 (0.289)	−0.148 (0.067)	0.061 (0.269)
Tech/2-Year	0.86 (0.16)	−0.25 (0.38)	−1.04 (0.04)				
College+	1.13 (0.22)	−0.00 (0.38)	−0.96 (0.04)				
SEP	−0.33 (0.08)	−1.47 (0.31)	−0.06 (0.02)	κ_{SEP}	<i>Panel C: Scale Parameters</i>		
				κ_{nonSEP}	1.410 (0.291)		
					1.258 (0.492)		
Rand. coeff. (σ_q)	0.85 (0.31)						

Notes: Panel A reports baseline demand parameters from Neilson (2026); the first quality entry is the common coefficient $\bar{\beta}^q$, subsequent education entries are deviations from the less-than-high-school category, and the SEP row is an additional shift. Price coefficients are signed. Panel B reports treatment-effect parameters estimated here using the sample in subsection 5.5.3. Panel C reports scale parameters and their standard errors. Standard errors in parentheses are clustered at the preschool-center level. $\psi_{\text{SEP}}^{\text{op}}$ is set to zero in the benchmark because the prevalence of zero out-of-pocket prices among SEP choices leaves this shift poorly identified (see Online Appendix for the unrestricted specification).

represented. The exercise therefore changes the weights and choice problems assigned to family types observed in the experimental population; it does not extend the treatment to the two higher-education types absent from the policy counterfactual. The maintained transport assumption is that the demand shift estimated within SEP status applies to families of the same status in the target markets. We refer the reader to Neilson (2026) for a thorough discussion of the baseline demand estimates and focus here on the new treatment parameters.

The main demand-side result is that the information treatment increases the weight families place on school quality. This effect is clearest for SEP families, where the estimated shift is economically meaningful relative to the baseline quality coefficient, but the pattern is similar for non-SEP families as well. In other words, the treatment does not simply move families mechanically toward specific schools; it makes school quality matter more in their choices.

The other treatment parameters point in the same direction. For SEP families, treatment raises the quality weight relative to the magnitude of the distance coefficient, making higher quality more attractive relative to proximity. For non-SEP families, the treatment makes out-of-pocket price matter somewhat more. Taken together, these estimates line up with the descriptive experimental evidence from Section 4: treated families place more weight on academic quality and are more willing to trade off convenience, and in some cases price, when choosing schools. Zero out-of-pocket prices at SEP schools limit the information available to estimate the SEP price shift.



Notes: Panel A: distribution of the estimated school-specific term $\hat{c}_j$ for voucher schools; dashed line marks the mean. Panels B and C: bin-averaged $\hat{c}_j$ against per-student expenditure and average teacher PSU scores for voucher schools. The term $\hat{c}_j$ may reflect persistent differences in both input costs and schools' objective functions. 22,882 school-year observations (2,389 schools), 53 markets, 2006–2015. Full parameter estimates in Table O-14.

The scale estimates provide only modest evidence that the intervention changed the overall sharpness of choice behavior. Both κ estimates are above one, but neither is statistically distinguishable from one at conventional two-sided levels under the displayed standard errors. The broader message is therefore not that the treatment primarily worked by dramatically reducing global choice noise. Rather, the data suggest that the main action comes through the characteristic-specific shifts, especially the increased weight on quality. Allowing for the scaled-logit possibility emphasized by Brown and Jeon (2024) is still useful conceptually, but in this application it does not materially overturn the main demand-side interpretation.

5.7. Supply Model Estimates

Figure 2 presents the supply-side parameter estimates from the two-regime GMM procedure described in subsubsection 5.5.5, estimated on voucher and private non-voucher schools over 10 years (2006–2015) across 53 markets.¹⁷ The estimation concentrates out school-specific terms c_j and market-year-type effects $\gamma_{mt}^{\text{type}}$, with type-specific time trends $\delta_{\text{type}} \cdot t$ allowing these terms to evolve differently across school types. Identification uses the policy and local-labor-market variation from the baseline demand instrument set, interactions with for-profit and nonprofit-voucher status, and a new Nearby Teacher Quality Index (Table O-15).

¹⁷The demand parameters are estimated on 2006, 2007, 2011, and 2012 as described in subsubsection 5.5.2. The extended supply panel uses the same demand estimates and recovers mean utilities for the additional years (2008–2010, 2013–2015) via BLP contraction mapping. Public schools are excluded from the main specification because their quality investment decisions are governed by municipal budgeting processes rather than the school-level optimality conditions in the model.

The supply estimates indicate substantial heterogeneity across schools, and the two-regime specification is empirically relevant rather than merely conceptual: 10.5% of school-year observations are classified as constrained and therefore governed by the composition-channel first-order condition rather than the standard enrollment-based one (subsubsection 5.5.5). The time trends by school type, reported with the full parameter estimates in Table O-14, are comparatively modest, so the main variation of interest comes from cross-school differences in competitive incentives and the school-specific terms in the quality FOC.

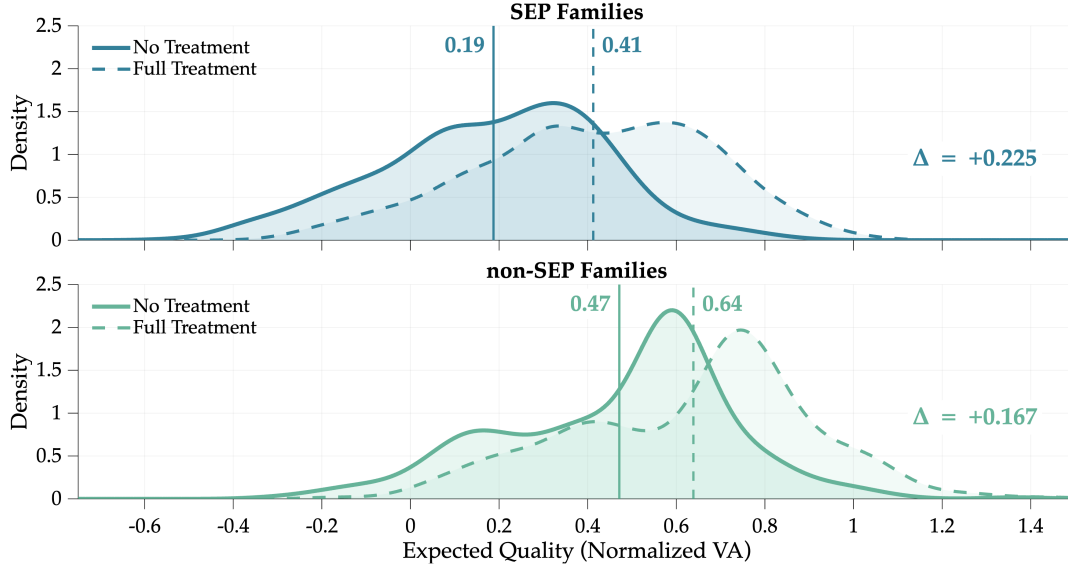
The relationship between competitive incentives and observed quality also has direct historical support. Using the SEP reform, Neilson (2026) relates changes in school value added to the model-implied pre- and post-reform quality markdowns and to changes in marginal revenue. The estimated coefficients on the two markdown terms are 0.96 and -0.98 , close to the values implied by the quality FOC, and the coefficient on the change in marginal revenue is 0.11 (s.e. 0.02). Our supply estimation uses the same economic relationship while allowing school-specific composite cost terms, sector-specific trends, capacity constraints, and the additional excluded variation described above.

Panel A of Figure 2 shows the distribution of the estimated school-specific terms for voucher schools. The distribution is concentrated at low values, with a mean of 0.35 and a tight dispersion. As discussed in subsubsection 5.4.1, these terms may reflect persistent differences in both input costs and schools' objective functions; they should not be interpreted as deep technological parameters. Panels B and C show that higher values of $\hat{c}_j$ are associated with greater expenditure per student and teachers with stronger observed credentials.¹⁸ These descriptive relationships are consistent with an input-cost component in $\hat{c}_j$, but they do not separately identify production technology and schools' objective functions.

In the next section, we use these demand and supply estimates to study how the policy would operate at scale under alternative assumptions about assignment, capacity, and equilibrium responses.

¹⁸Per-student expenditure comes from MINEDUC's accounting data (*Gasto Rendido*) and includes both personnel and operational costs.

FIGURE 3.—Counterfactual: Quality Distribution under Full Take-up (Experiment Sample)



Notes: Distribution of expected school quality (normalized VA) for experiment-sample families. The counterfactual treats eligible families (types 1–4) at full take-up ($\tau = 1$), holding school quality and prices fixed. Top panel: SEP families. Bottom panel: non-SEP families. Solid lines: baseline (no treatment). Dashed lines: full treatment. Vertical lines mark group means. The curves use the restricted treatment specification carried into the equilibrium counterfactuals.

6. EQUILIBRIUM EFFECTS

6.1. Counterfactual Policy Simulations

We now use the estimated model to ask what would happen if the information intervention were implemented at scale rather than in a small experiment. We build up the answer in layers, isolating the contribution of each equilibrium force. The first benchmark holds school characteristics fixed and asks how demand shifts alone would change the distribution of school quality. The second introduces capacity constraints through centralized assignment. The third allows schools to adjust quality in response to the changed competitive environment. This layered approach makes clear which forces matter and in which direction.

Formally, let $\tau_{nk} \in [0, 1]$ denote the fraction of type- k families at residential location n who receive the treatment, following the spatial structure of Neilson (2026). Each family's school choice probabilities are a mixture of the treated and control predictions from the demand model. Aggregating these mixed shares across locations yields market-level demand by type; the average quality attended by each type is the corresponding share-weighted mean over schools.

Model Fit and At-Scale Predictions Before exploring the layered counterfactuals, we verify that the structural model reproduces the experimental evidence and then examine how the

TABLE V
SUMMARY OF DEMAND-SIDE POLICY EFFECTS

	<i>Panel A: Treatment Effects on Quality</i>			<i>Panel B: Quality Gap</i>	
	Experiment	Model (demand only)		No treatment	−0.214
	RCT ATE	Exp. sample	All mkts	Full treatment	−0.173
SEP families	0.220 (0.087)	0.221 (0.081)	0.186 (0.075)	Narrowing	0.041 (19%)
Non-SEP families	0.114 (0.104)	0.151 (0.099)	0.144 (0.085)		

Notes: Panel A reports *treatment effects*—the difference in mean school quality between full treatment ($\tau = 1$) and no treatment ($\tau = 0$), holding school characteristics fixed. Column 1 is the experimental RCT ATE from Table III. Column 2 is the model prediction in the experiment-sample markets. Column 3 is the model prediction across all 53 urban markets; this is the demand-only benchmark carried forward in Table VI, Column (2). Panel B reports the SEP–non-SEP quality gap under no treatment and full treatment across all markets. Quality is measured in standard deviations of normalized value added. Standard errors are in parentheses. Column (1) uses the preschool-clustered experimental standard error. For Columns (2)–(3), numerical delta-method standard errors propagate the preschool-clustered joint covariance of the estimated experimental treatment parameters through the model prediction. Column (2), an in-sample fit check, is evaluated at the unrestricted treatment estimates and propagates all eight parameters. Column (3) is evaluated at the restricted estimates imposing $\psi_{SEP}^{op} = 0$, the specification used in the equilibrium counterfactuals, and propagates the seven free parameters. Columns (2)–(3) involve no assignment simulation; each perturbation re-evaluates the model’s choice probabilities and expected qualities. The model standard errors are conditional on the demand parameters drawn from Neilson (2026) and the maintained model structure.

implied treatment effect changes when extrapolated to the broader population. Table V reports three objects for each family group. Column 1 is the experimental RCT average treatment effect on school quality from Table III. Column 2 uses the unrestricted treatment estimates to compute the model-predicted effect over the same experiment-sample markets—an in-sample fit check confirming that the structural parameters reproduce the targeted moments when aggregated through the counterfactual machinery. Column 3 uses the restricted treatment specification, imposing $\psi_{SEP}^{op} = 0$, to compute the effect over all 53 urban markets in the model population. The comparison therefore reflects both the population change and the treatment-parameter restriction: for SEP families, the population effect (0.186 SD) is somewhat smaller than the experiment-sample prediction (0.221 SD). This is consistent with the experiment operating in markets where low-income families face larger quality gaps and therefore have more room to gain from information. The population extrapolation is the relevant benchmark for the policy simulations that follow.

At-Scale Demand Responses As a first benchmark, we ask what the treatment would imply if schools had not yet adjusted—the natural partial-equilibrium starting point. We apply the treatment to eligible families (types 1–4) at full take-up ($\tau = 1$) while holding school quality and prices fixed, and compute the implied distribution of school quality for SEP and non-SEP families (Figure 3). The demand shift produces a substantial rightward movement in quality for SEP families, reflecting the large quality weight increase estimated in Table IV.

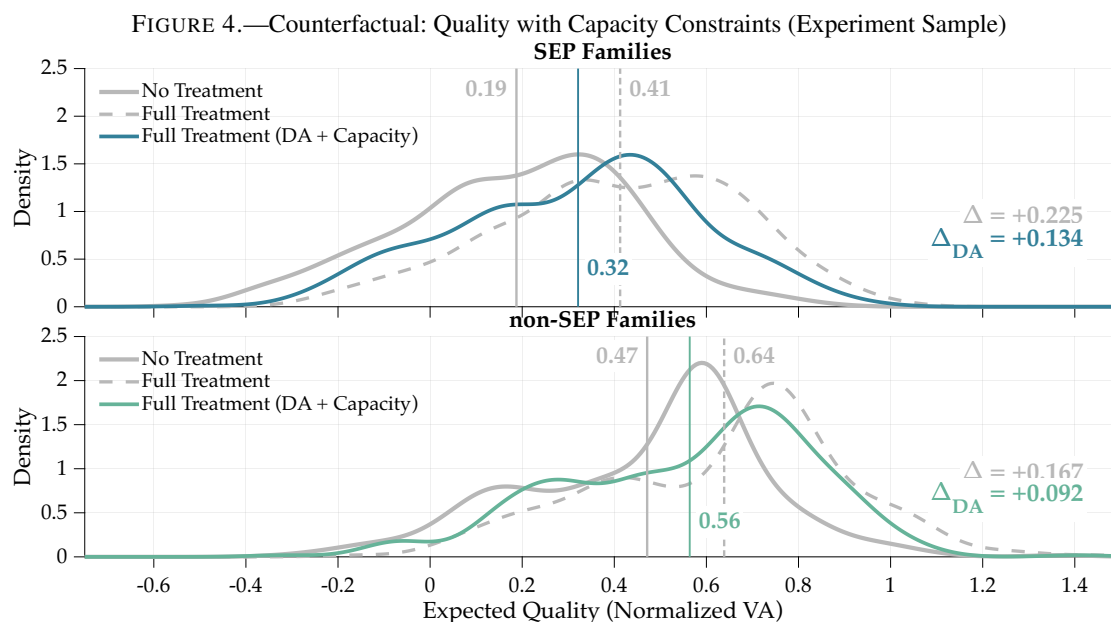
This benchmark holds school value added fixed as students re-sort. Because value added is measured at the observed student composition, it can contain a contribution from peers at that composition. The baseline model nevertheless does not allow changes in peer composition to affect our measure of school quality. The Online Appendix reports a sensitivity in which value added responds linearly to the change in a school’s non-SEP share. The model allows student composition to change and allows SEP composition to affect schools’ incentives through targeted voucher revenue; it does not include a separate family preference for peer attributes or peer composition in schools’ objectives.

However, these demand shifts imply that the most popular schools would have to expand enrollment substantially—a requirement that may not be feasible in the short run. This does not contradict the baseline demand approximation used by Neilson (2026), where capacity constraints are not a first-order concern in the historical data. The scale-up exercise asks a different question: once the information policy is applied broadly, demand can become sufficiently concentrated at popular schools that enrollment limits matter for realized assignments even if they were second-order in the estimation sample.

Capacity-Constrained Counterfactual The demand-only benchmark assumes every family can attend their most-preferred school. In practice, popular schools face enrollment limits. To assess how congestion attenuates treatment effects, we simulate a student-proposing Deferred Acceptance (DA) algorithm in each market, where families submit rank-ordered preference lists derived from their logit choice probabilities and schools rank applicants by random lottery priority, as in Chile’s centralized system.¹⁹

School capacities are imputed from administrative records. For public and voucher schools we use reported enrollment slots (*cupos totales*) from Chile’s 2020 school-level enrollment data (SAE), adjusted to the estimation period using historical classroom counts (see Online Ap-

¹⁹To generate rank-ordered preference lists consistent with the multinomial logit model, we draw $\tilde{U}_{ij} = V_{ij} + \varepsilon_{ij}$ for each family-school pair, with ε_{ij} i.i.d. Gumbel, and then rank schools by $\tilde{U}_{ij}$.



Notes: Gray solid: baseline. Gray dashed: full treatment, free choice. Colored solid: full treatment under DA with capacity constraints. Treatment applied to types 1–4; types 5–6 untreated. The curves use the restricted treatment specification carried into the equilibrium counterfactuals. Mean labels are rounded to two decimal places; the treatment effects printed in the figure are calculated from the unrounded means.

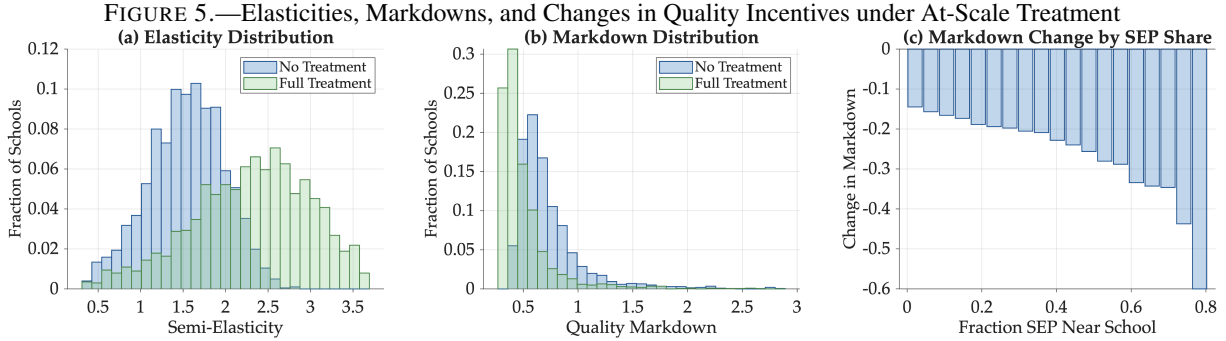
pendix O-14.2 for details). For unmatched and private schools we use alternative administrative capacity measures and market-level medians as fallbacks. The resulting enrollment-to-capacity ratio averages 0.73, indicating meaningful slack in most markets. The DA exercise should therefore be interpreted as a short-run congestion benchmark, not as evidence that household choices in the baseline estimation sample were broadly rationed.

Figure 4 plots the quality distributions under three scenarios: baseline, demand-only (free choice), and demand under DA with capacity constraints. Table VI decomposes the equilibrium effects by reporting *mean quality levels* across the population under each successive layer—the treatment effects in Panel B are the implied differences relative to the no-treatment baseline. Column (2) corresponds to the demand-only population benchmark from Table V, Column 3; the remaining columns progressively add capacity constraints and supply responses. Congestion matters: roughly half of the demand-only treatment effect survives capacity constraints (Column 3). Untreated higher-education families (types 5–6) experience a small negative displacement as treated families crowd into higher-quality schools—a spillover that the supply response in the next section can reverse.

TABLE VI
COUNTERFACTUAL SIMULATIONS: MEAN QUALITY AND TREATMENT EFFECTS (POPULATION)

	(1)	(2)	Equilibrium (DA + Capacity)		
			(3)	(4)	(5)
	Baseline	Demand Only	No Supply Response	Supply Response	Supply Resp. (+10% MC)
<i>Panel A: Mean Expected Quality</i>					
All (types 1–6)	0.558	0.680	0.619	0.723	0.615
SEP (types 1, 3)	0.298	0.483	0.407	0.526	0.444
Non-SEP (types 2, 4)	0.512	0.656	0.588	0.702	0.599
<i>Panel B: Treatment Effects (relative to Baseline)</i>					
All (types 1–6)		0.122	0.062	0.166	0.057
		(0.046)	(0.025)	(0.041)	
SEP (types 1, 3)		0.186	0.109	0.229	0.147
		(0.075)	(0.059)	(0.061)	
Non-SEP (types 2, 4)		0.144	0.076	0.191	0.087
		(0.085)	(0.059)	(0.076)	
<i>Panel C: Differences Across Equilibrium Concepts</i>					
	All	SEP	Non-SEP		
Congestion attenuation, (2) – (3)	0.060	0.077	0.068		
	(0.027)	(0.031)	(0.034)		
Supply response, (4) – (3)	0.104	0.120	0.115		
	(0.040)	(0.040)	(0.044)		
Net equilibrium correction, (4) – (2)	0.044	0.043	0.046		
	(0.031)	(0.034)	(0.032)		

Notes: Panel A reports population-weighted mean expected quality across all markets; all-family averages include types 1–6, while treatment applies to types 1–4; Panel B reports the implied treatment effects (each column minus Column 1); Panel C reports differences across equilibrium concepts. Column (2) is the demand-only benchmark from Table V, Column 3, reported here as a level to facilitate comparison with the equilibrium columns. Columns (3)–(5) impose deferred acceptance with administrative capacity constraints. “No Supply Response”: schools do not adjust quality. “Supply Response”: schools adjust quality via the estimated FOC; public schools held fixed. Column (5) scales all schools’ unit cost of quality c_j by 1.10 before solving for the equilibrium, capturing potential upward pressure on input costs when many schools improve quality simultaneously (Tincani, 2021). Capacities come from administrative data (Online Appendix O-14.2). Numerical delta-method standard errors in parentheses propagate the preschool-clustered joint covariance of the seven estimated experimental treatment parameters through the full assignment and supply-equilibrium mapping. Each perturbation re-solves the counterfactual with common simulated family draws and assignment priorities. These standard errors are conditional on the baseline demand parameters taken from Neilson (2026), estimated supply inputs, administrative capacities, and maintained counterfactual structure. Uncertainty about these objects is examined through the separate specification and sensitivity exercises. Column (5) reports point estimates only for the cost sensitivity.



Notes: Demand-side only (no supply adjustment), eligible families (types 1–4) treated at full take-up ($\tau = 1$). Panel (a): distribution of the quality semi-elasticity of demand $(\partial s_j / \partial q_j) / s_j$ for voucher schools under baseline and full treatment. Panel (b): distribution of quality markdowns $s_j / (\partial s_j / \partial q_j)$ under baseline and full treatment. Panel (c): median change in markdown by the fraction of SEP-eligible students near the school. 53 urban markets.

6.2. Supply Responses

The capacity-constrained simulations show that congestion attenuates roughly half of the demand-side gains when the policy is scaled up. The central question is whether supply-side responses—schools adjusting quality in response to stronger competitive pressure—can offset this attenuation. We interpret the exercise as a medium-run approximation: schools adjust quality given the shifted demand system and observed capacity, but we abstract from entry, exit, and capacity investment.

Before computing the equilibrium, it is useful to see how the information treatment changes competitive incentives on impact. Figure 5 presents three panels. Panel (a) shows the distribution of the quality semi-elasticity of demand under baseline and full treatment: the median increases from 1.55 to 2.37, a 53% increase. Panel (b) shows the corresponding quality markdowns, which compress from a median of 0.65 to 0.42—a 35% reduction in market power over quality. Panel (c) shows that this compression is largest for schools in neighborhoods with the highest concentration of SEP-eligible students, consistent with the treatment having the greatest impact on the families that were previously least informed. The policy therefore does not just shift demand; it also changes the competitive environment in which schools operate, especially in the neighborhoods where baseline inequality is greatest.

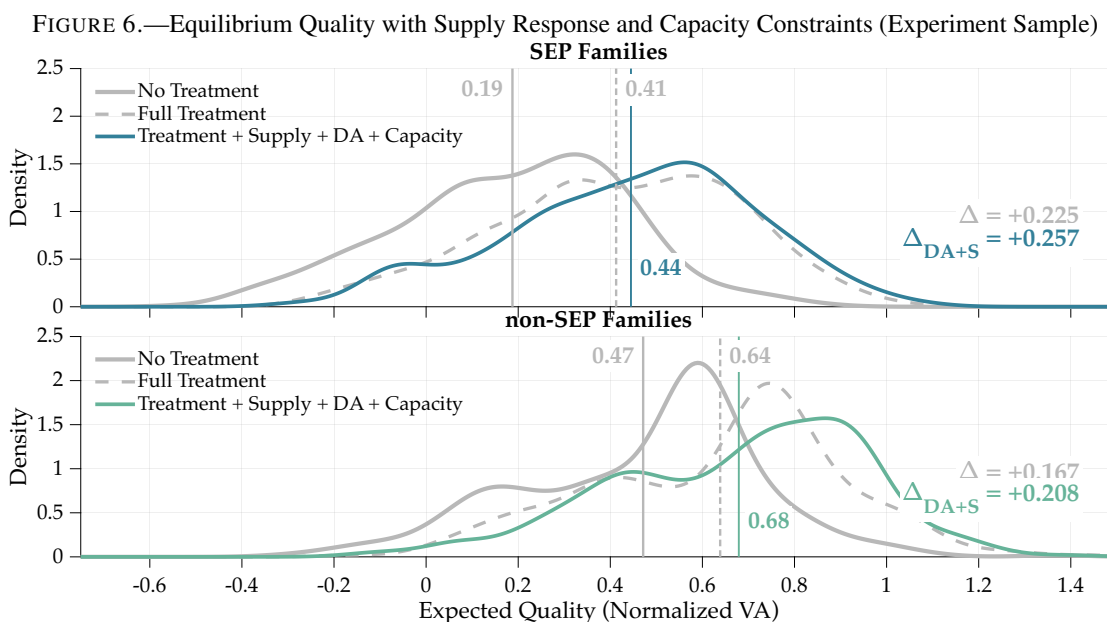
These changes in competitive incentives motivate the equilibrium exercise. We solve each school's quality FOC in changes around the observed equilibrium, holding fixed its school-specific unobservable. For unconstrained schools, the equilibrium update simplifies to $Q^{\text{eq}} =$

$Q^{\text{data}} + \text{MD}^{\text{base}} - \text{MD}^{\text{treat}}(Q^{\text{eq}})$, so the quality response is governed by how markdowns change. For schools near capacity, the composition-channel FOC (Equation 11) instead links quality adjustment to the policy-induced change in the marginal return from attracting SEP students. The historical supply estimates characterize heterogeneity in the input-cost and objective-function terms underlying these conditions; the robustness exercise below examines how the equilibrium changes when the implied marginal cost of quality is increased.

We iterate the two-regime fixed-point equation—updating demand, recomputing markdowns and composition-channel derivatives, and adjusting quality with damping—until convergence. Capacity status is classified dynamically at each iteration based on implied enrollment. Schools facing larger reductions in markdowns increase quality more. Public schools are held fixed ($\Delta q_j = 0$). As discussed in subsubsection 5.4.1, this is a scope restriction motivated by municipal budgeting and staffing institutions; its sign is not assumed to be conservative.

Figure 6 shows the result. The supply response substantially offsets the attenuation from capacity constraints, bringing the equilibrium distribution back toward the demand-only benchmark. For SEP families in the experiment sample, the combined demand-plus-supply treatment effect under DA is $\Delta_{DA+S} = +0.257$ SD, compared to $\Delta_{DA} = +0.134$ with demand only under DA and $\Delta = +0.225$ under free choice (restricted specification, Figure 5). Non-SEP families experience similar patterns ($\Delta_{DA+S} = +0.208$ versus $\Delta_{DA} = +0.092$). At the population level (Table VI), congestion reduces the demand-only effect, while the supply response significantly offsets that reduction. The final equilibrium point estimate is slightly above, but not statistically distinguishable from, the demand-only benchmark.

We also quantify how much unmodeled input-price pressure would be required to overturn this result. The appendix raises the composite marginal cost of quality uniformly for all private schools and re-solves the equilibrium. A 10% increase reduces the all-family effect from 0.166 to 0.057 SD, while a 20% increase reduces it to -0.034 SD. The corresponding SEP effect remains positive at 0.077 SD even in the 20% scenario. Thus, under this convention, eliminating the overall effect requires a cost increase between 10% and 20%. Chile’s regulated municipal-school pay schedules may anchor wages in the broader teacher market, although private voucher schools retain flexibility over supplemental pay and can recruit across sectors (Bertoni et al., 2018). Because the exact crossing point depends on the location normalization of the quality index, we interpret this as a transparent sensitivity range rather than a normalization-invariant structural threshold.



Notes: Distribution of expected school quality (normalized VA) for experiment-sample families under three scenarios. Gray solid: baseline (no treatment). Gray dashed: full treatment, demand only (no supply response). Colored solid: full treatment under deferred acceptance with administrative capacity constraints and supply-side quality response. Top panel: SEP families. Bottom panel: non-SEP families. Public schools held fixed ($\Delta q = 0$). The curves use the restricted treatment specification carried into the equilibrium counterfactuals.

The central lesson is that partial-equilibrium demand calculations can mislead in opposite directions depending on which equilibrium margins are omitted. A naive extrapolation that ignores congestion would overstate the short-run gains. But an analysis that accounts for congestion without allowing supply responses would understate the medium-run gains. In the full equilibrium, the competitive channel—schools facing stronger incentives to improve quality—largely offsets the congestion channel, returning the effect close to the demand-only benchmark.

6.3. Robustness and Discussion

The equilibrium exercises rest on several modeling choices and institutional assumptions. We discuss the most important ones here; Table O-17 in the appendix reports additional robustness exercises—including targeted treatment of only SEP or non-SEP families, allowing public schools to respond, and partial take-up—which confirm that the main conclusions are stable and that supply-side quality improvements generate positive spillovers for untreated families.

On the household side, the demand system is static: it represents the choice problem faced by one entering cohort and holds prior sibling placements and family composition fixed. It therefore does not separately model the sequence of school choices across children within a family.

The baseline demand parameters are estimated from realized population choices and thus reflect the average influence of persistent family constraints, although sibling-school preferences are not separately identified. The data directly bound the composition margin relevant to transporting the experimental demand shift: older-sibling prevalence is similar across the experimental timing groups, the choice response is not detectably attenuated among sibling families, and adding sibling status leaves the experimental moments essentially unchanged. At first appearance in the eventual primary school, 31.9% of the structural sample and 37.1% of comparable families in the model markets have an older sibling in grades 1–8; among them, 72.5% and 75.6%, respectively, enter the sibling’s school (see the Online Appendix). The counterfactuals should nevertheless be read as effects for a single entering cohort, not as a model of how the intervention changes younger siblings’ future choices.

Over a longer horizon, entry, exit, or capacity investment could in principle respond to a sustained demand shift. In practice, prior work in the Chilean context finds that these margins have not been large drivers of market-level change, given the existing market structure and apparent excess capacity (Neilson, 2026). Similar to the analysis in Wollmann (2018), our counterfactual focuses on the adjustment of product characteristics by incumbent schools. We do not model the direction or magnitude of longer-run entry, exit, and capacity responses.

The counterfactual also leans on two institutional features of the Chilean environment. First, restrictions on school selection predate the current assignment mechanism: SEP prohibited participating schools from screening applicants, and the 2009 General Education Law extended the prohibition more broadly across subsidized elementary schools. The later centralized deferred-acceptance system makes applications and the allocation of oversubscribed seats more transparent: families rank multiple schools, and assignments follow centrally specified priorities and lottery numbers. This does not hold student composition fixed; composition can still change when the information policy changes applications and the resulting assignments. Second, voucher-school copayments were heavily regulated during our estimation period; the 2015 Inclusion Law subsequently froze existing copayments and began phasing out shared financing.²⁰ These institutions motivate a medium-run benchmark with quality as the school adjustment margin. We hold observed tuition fixed in all sectors; for private non-voucher schools, this is a maintained modeling assumption.

²⁰Even before the 2015 law, the shared-financing rules (DFL No. 2 of 1998) capped copayments and reduced the general subsidy as fees rose; SEP required participating schools to charge eligible students zero. See Sanchez (2017) for related discussion.

7. DESIGNING A LARGE-SCALE RCT

The preceding sections established that equilibrium forces—congestion and supply-side quality adjustment—are quantitatively important for evaluating the information policy at scale, and that an individual-level RCT cannot capture them: if treating one family shifts demand at a school, nearby control families are affected too. Cluster-randomized trials—which assign entire geographic areas to treatment or control—can recover such effects by comparing outcomes across differentially saturated clusters (e.g., Miguel and Kremer, 2004, Muralidharan and Sundararaman, 2015). In school choice, the natural cluster would be the market, but with only 53 urban markets available in this setting, market-level randomization may lack statistical power. The model, however, reveals that competitive pressure is highly location-specific: a school’s demand depends primarily on nearby families, and the supply response depends on local mark-down compression. This locality means that spillovers decay with distance, making it possible to create geographic clusters *within* a single market—small enough to yield many independent units for estimation, yet large enough to contain the relevant competitive interactions. The remaining design problem is contamination: equilibrium responses propagate through competition among geographically connected schools, so schools near cluster boundaries are not independent. At the same time, having more clusters improves statistical power.

We use the estimated model to evaluate candidate cluster-randomized designs and study their statistical properties before the experiment is fielded. We focus on Santiago, the market where the original experiment was conducted (Figure 7, Panel a), and study how cluster size, treated area, and buffer zones shape the bias and precision of the design.

Design. We tile Santiago with a regular hexagonal grid of circumradius r km (Figure 7).²¹ Each eligible hexagon is assigned to treatment or control via stratified randomization, balanced 50/50 within strata defined by distance from the market centroid and angular quadrant.

Within each hexagon, two concentric zones serve different roles. The *core* (distance $< I$ km from center) is the unit of observation: only schools in the core zone enter the regression. The surrounding *treatment ring* (distance I to r) extends the treated area so that core schools face high demand saturation from treated neighbors, but ring schools are excluded from the regres-

²¹Hexagons approximate circles—minimizing the boundary-to-area ratio—while tiling the plane without gaps, making them a natural choice for spatial cluster designs.

sion to avoid noisy near-boundary observations. In treated hexagons, one quarter of families receive the information intervention ($\tau = 0.25$); in control hexagons no one does.²²

Estimand and estimator. The estimand is $\beta^* = \bar{Y}(\tau=1) - \bar{Y}(\tau=0)$, the average treatment effect on school quality at the specified take-up rate. The outcome is the school-level quality change induced by the supply-side response, ΔQ_j , for voucher and private schools in the core zone—the schools whose quality adjusts via the supply-side FOC.²³ The estimator is

$$\Delta Q_j = \alpha + \beta T_{h(j)} + \gamma Q_{j,\text{baseline}} + \epsilon_j, \quad (14)$$

where ΔQ_j is the quality change at school j in the core of hexagon h , $T_{h(j)}$ is the treatment indicator for that hexagon, and $Q_{j,\text{baseline}}$ is pre-treatment quality. Standard errors are clustered at the hexagon level.²⁴

Bias–power trade-off. The fundamental trade-off is between contamination bias and statistical power. Smaller clusters share more schools across treatment-control boundaries (increasing the potential for contamination and downward bias), while larger clusters leave fewer independent units for estimation (decreasing power). Because treatment and control clusters share schools near their boundaries, the estimator is attenuated. A useful heuristic is the exposure contrast $\rho = \bar{\lambda}^T - \bar{\lambda}^C$, where λ_j is the treated share of school j ’s demand: in a simple difference-in-means estimator, bias would be approximately $-(1 - \rho)\beta^*$. In practice, the three-zone design reduces contamination well below this bound by excluding near-boundary schools from the regression. We optimize directly over the simulation-based RMSE, computed across stratified random assignments rather than inferred from the analytic approximation.²⁵

Simulation. We evaluate a grid of cluster designs, varying the hex radius from 0.75 to 4 kilometers and the core radius from 0.5 to 2 kilometers. For each configuration, we run multiple simulation replicates—each drawing structural parameters from their asymptotic distribution,

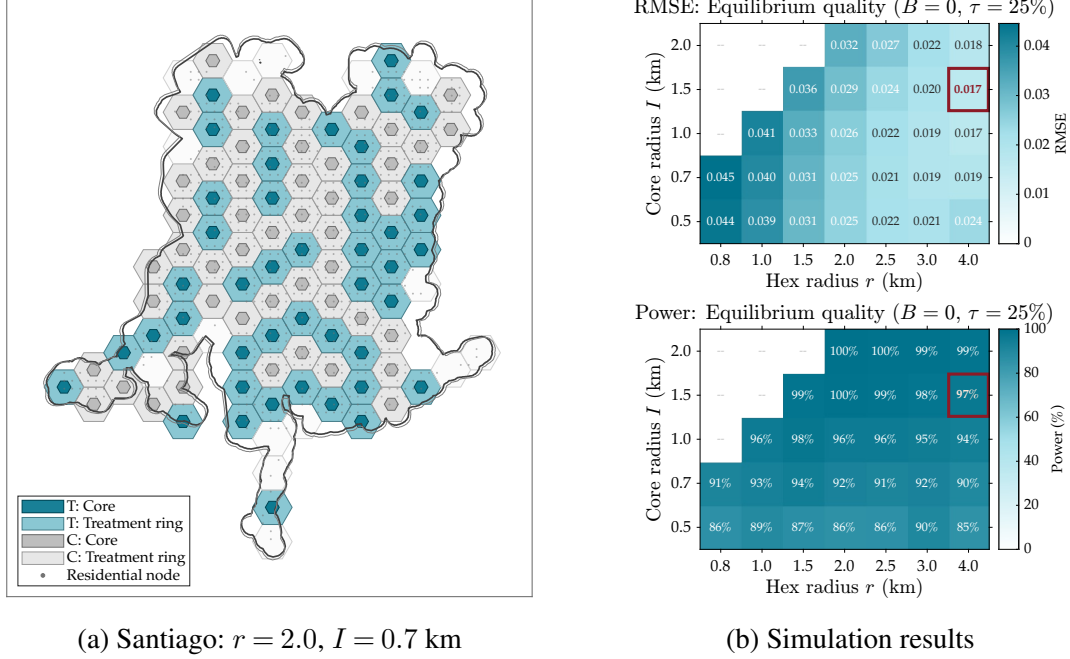
²²The appendix examines sensitivity to higher take-up rates.

²³Public schools are excluded from the regression because they do not respond to competitive incentives through the model’s supply channel.

²⁴Because the design is restricted to Santiago, the regression does not include market fixed effects. In a multi-market implementation, market fixed effects would absorb between-market variation in school quality.

²⁵Each simulation replicate draws treatment parameters from their asymptotic distribution, so that the reported RMSE and power account for estimation uncertainty in the structural model as well as randomization variance.

FIGURE 7.—Scale-Up RCT Design



Notes: Panel (a): one stratified random assignment of hexagonal clusters to treatment (teal) and control (gray) in Santiago. Dark inner hexagons are the core zone; lighter outer zones are the treatment ring. Panel (b): RMSE and power for each (r, I) configuration. Highlighted cell minimizes RMSE subject to power $\geq 80\%$. School-level regression on ΔQ_j for voucher and private schools, hex-clustered SEs, $\tau = 0.25$, and no buffer strip ($B = 0$).

a random hex grid origin (to average over grid placement), and a stratified hex assignment—and estimate (14) on Santiago clusters. The outcome is the equilibrium quality response: we solve the supply-side FOC under partial treatment and measure how much each school’s quality changes relative to the no-treatment baseline ($\beta^* \approx 0.05$ s.d. at $\tau = 0.25$).²⁶

Figure 7 presents the results. Panel (a) shows one realization of the hexagonal design overlaid on Santiago ($r = 2.0$ km, $I = 0.7$ km). Teal hexagons are treated clusters, gray are control; darker inner hexagons mark the core zone. Panel (b) shows RMSE and power heatmaps over the design space, with the constrained optimum highlighted.

Results. The two panels of Figure 7(b) reveal a clear bias–power trade-off. RMSE decreases monotonically with hex radius r : larger clusters place core schools farther from hex boundaries, reducing contamination from cross-boundary demand spillovers. Power increases with core radius I : a wider core includes more schools in the regression, shrinking standard errors. The

²⁶The appendix also reports results for the demand-only outcome (expected school quality without supply adjustment). That estimand is larger, and power is adequate for most designs, so the bias–power trade-off is less informative as a design tool.

tension is that the lowest-RMSE configurations (large r , small I) tend to have low power, because a tiny core inside a large hex leaves few schools for estimation.

The constrained optimum—the design that minimizes RMSE subject to power $\geq 80\%$ —is $r = 4.0$ km, $I = 1.5$ km (RMSE = 0.017, power = 97%). More conservative choices such as $r = 2.0$ km, $I = 1.0$ km achieve similar power (96%) at somewhat higher RMSE (0.026). Across the grid, well-powered designs have RMSE of at most about 0.045, indicating that the equilibrium quality effect ($\beta^* \approx 0.05$ s.d.) is detectable under a range of practical configurations.

The model thus provides concrete guidance for the scale-up design: the locality of competitive interactions—the fact that demand and supply spillovers decay with distance—makes within-market cluster randomization feasible, and the simulation grid identifies the configurations that balance bias against power. A researcher would choose among the well-powered designs based on operational constraints (e.g., how many hexagons are administratively manageable) and the relative weight placed on bias versus precision.

8. DISCUSSION

This paper studies how a low-cost information policy affects school choice and what its effects would be at scale. In the experiment, treated families—especially those reached before they had enrolled—choose schools with higher test scores and value added, and their children score higher in fourth grade. Estimates remain positive in later observed assessment samples, although those samples are smaller and selected by test participation. Together, these results show that a simple intervention delivered through existing preschool infrastructure can shift actual choices toward higher-quality schools and improve academic outcomes.

The policy question, however, is not only whether the intervention works in a small-scale trial, but what happens once it is implemented broadly enough to change market conditions. Our counterfactuals show that this extrapolation is not mechanical. Capacity constraints attenuate the demand-side gains due to congestion when more families seek access to the same high-quality schools. At the same time, the increase in demand for quality changes schools' incentives, and in our preferred specifications the resulting supply-side quality response largely offsets the attenuation from congestion. The quantitative effects depend on assumptions about assignment, take-up, cost pressure, and which schools adjust. Across these exercises, however, the broader lesson is stable: equilibrium forces are central for evaluating a demand-shifting policy in a school market.

The methodological contribution comes from the complementary roles of the experiment and the model. We do not rely on the structural model alone to infer the demand shift induced by the policy; the randomized trial provides that object directly and credibly. But no small-scale experiment can identify the equilibrium effects that arise only when enough families are treated to change congestion, assignment outcomes, and schools' quality incentives. The model provides that equilibrium extension. It also allows us to study policy margins that are directly relevant for implementation but cannot be learned from the experiment alone, including alternative targeting rules, partial take-up, spillovers across family types, and different assumptions about how schools respond. In that sense, the framework turns a small-scale experiment into a disciplined tool for prospective policy analysis rather than treating it as a stand-alone estimate of policy impact.

The same framework is useful for experimental design. When a policy is rolled out gradually or across only part of a set of urban markets, the design of the larger-scale evaluation determines what can be learned about equilibrium effects. Our scale-up exercise shows how the estimated model can be used to choose clusters and buffer zones in a way that balances statistical power against contamination from spillovers. More broadly, the paper illustrates how a credible small-scale experiment, combined with enough economic structure to analyze market adjustment, can provide disciplined guidance about both the likely equilibrium effects of a new policy and the design of the larger-scale experiments needed to learn more about it.

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APPENDIX

A1. *Reduced-Form Regression Specifications*

Tables I and II report intent-to-treat coefficients from reduced-form regressions of the form

$$Y_i = \alpha + \beta T_p + Z_p' \delta + X_i' \gamma + \mu_c + \varepsilon_i,$$

where T_p is treatment assignment at the preschool-center level, μ_c are preschool-municipality fixed effects, and standard errors are clustered at the preschool-center level.

In the main specification, the market controls are nine market-level variables: the average SIMCE score used in the report card, the number of schools in the market, the 10th, 25th, 50th, 75th, and 90th percentiles of the SIMCE distribution, the interquartile range of SIMCE scores, and the standard deviation of SIMCE scores. The expanded family controls are child sex, SEP priority eligibility, indicators for the mother having completed high school or higher education, maternal marital status, household size, an index of durable-goods ownership, and home ownership. Missing-value indicators are included for every family covariate and missing values are set to zero, so the controls do not alter the estimation samples.

The Online Appendix reports estimates with no family controls, maternal education only, alternative birth-record controls, and richer combinations of demographic and survey controls. These controls are introduced as complete blocks rather than selected separately by outcome. It also reports balance and coverage for the controls used in the main specification. The household variables are observed for every child in Panels B and C, apart from one missing home-ownership value in Panel C; missing-indicator coding preserves the larger pooled sample.

Panel A uses the full experimental sample. Panel B restricts to families who had already enrolled at the time of the Pre-K visit, and Panel C restricts to families who had not yet enrolled. For Table I, column 1 restricts the distance outcome to observations with distance below 4 km. Columns 2–6 use the subsample with non-missing school value added so that the school-choice outcomes are compared on a common sample. For Table II, columns 7–8 are conditional on taking the college entrance exam, and the reported scores pool the 2023–2025 cohorts standardized within cohort.

A2. *Experiment Moments for Structural Estimation*

Table III reports the experimental moments used in the second-step demand estimation. These regressions use the model-overlap sample described in Section 5.5.3: families who had not yet enrolled at the time of treatment, can be matched to the six markets covered by the baseline demand model, pass the geographic consistency filter, and have non-missing school characteristics used in estimation.

The structural moment regressions include preschool-municipality fixed effects, the nine market-level SIMCE controls listed in Appendix A1, indicators for the mother having completed high school or higher education, and a missing-education flag. Standard errors are clustered at the preschool-center level. This narrow maternal-education block differs from the expanded family controls in the reduced-form analysis. The GMM vector stacks effects on five chosen-school outcomes—quality (normalized value added Q_2), sticker price, out-of-pocket price, Google Maps driving distance restricted to observations within 4 km, and voucher-school attendance—separately for SEP and non-SEP families. These ten entries are followed by two residualized mean-utility entries, one per group, giving twelve entries in total. Pooled estimates are descriptive and do not add entries to the GMM objective.

A3. *Scaled-Logit Derivation*

This subsection gives a simplified type-level illustration of the scaled-logit interpretation summarized in Section 5.3, suppressing within-type random quality tastes for exposition. The purpose is interpretive: it shows why imperfect information about school characteristics can motivate both characteristic-specific treatment effects and a common scale shift in the choice model. The demand estimation and counterfactual exercises in the main text are implemented directly in terms of the effective parameters, so none of the policy results requires separately identifying the underlying signal process.

Suppose that for each characteristic $c \in \{q, \text{op}, d\}$, family i of type k observes a noisy signal

$$\tilde{s}_{ij}^c = c_j + v_{ij}^c,$$

where $v_{ij}^c \sim N(0, \sigma_{v,c,k}^2)$ is independent noise. Given a prior $c_j \sim N(0, \sigma_c^2)$, the posterior mean is

$$E[c_j \mid \tilde{s}_{ij}^c] = \rho_k^c \cdot \tilde{s}_{ij}^c, \quad \rho_k^c = \frac{\sigma_c^2}{\sigma_c^2 + \sigma_{v,c,k}^2} \in (0, 1), \quad (15)$$

where ρ_k^c is the signal-to-noise ratio for characteristic c and type k .

Under this simplification, substituting the posterior means into utility yields expected utility of the form

$$U_{ij}^{\mathcal{E}} = \phi_k^q q_j + \phi_k^{op} \text{op}_{k(i),j} + \phi_k^d d_{\text{loc}(i),j} + \bar{\beta} x_j + \xi_j + \tilde{\epsilon}_{ij}, \quad (16)$$

where $\phi_k^q = \beta_k \rho_k^q$, $\phi_k^{op} = \alpha_k \rho_k^{op}$, and $\phi_k^d = \lambda_k \rho_k^d$ are products of structural preferences and signal precision. Thus, noisier signals attenuate the effective weight placed on the corresponding characteristic. The composite error is

$$\tilde{\epsilon}_{ij} = \phi_k^q v_{ij}^q + \phi_k^{op} v_{ij}^{op} + \phi_k^d v_{ij}^d + \epsilon_{ij}. \quad (17)$$

The variance of this composite error is

$$\text{Var}(\tilde{\epsilon}_{ij} \mid k) = (\phi_k^q)^2 \sigma_{v,q,k}^2 + (\phi_k^{op})^2 \sigma_{v,op,k}^2 + (\phi_k^d)^2 \sigma_{v,d,k}^2 + \frac{\pi^2}{6}. \quad (18)$$

As in Brown and Jeon (2024), improving signal precision can therefore affect not only the effective coefficients on informed characteristics but also the overall error variance of the choice model.

We approximate this environment with a scaled logit:

$$\tilde{\epsilon}_{ij} \approx \mu_k \cdot \epsilon_{ij}^*,$$

where ϵ_{ij}^* is i.i.d. EV1 and the type-specific scale is chosen to match the variance of the composite error:

$$\mu_k = \sqrt{1 + \frac{6 \sigma_{\text{noise},k}^2}{\pi^2}} \geq 1, \quad (19)$$

with

$$\sigma_{\text{noise},k}^2 \equiv (\phi_k^q)^2 \sigma_{v,q,k}^2 + (\phi_k^{op})^2 \sigma_{v,op,k}^2 + (\phi_k^d)^2 \sigma_{v,d,k}^2.$$

Dividing expected utility by μ_k then yields standard logit choice probabilities with effective parameters equal to structural parameters divided by the error scale.

The baseline demand estimation in Neilson (2026) identifies these effective parameters under the normalization $\mu_k^C = 1$. If treatment changes signal precision, the treated and control scales

need not coincide. The common scale parameter in Section 5.2 is therefore

$$\kappa_k \equiv \frac{\mu_k^C}{\mu_k^T}. \quad (20)$$

When improved signal precision reduces composite error variance, $\mu_k^T < \mu_k^C$ and hence $\kappa_k > 1$: choice behavior becomes sharper even for school attributes about which the intervention provides no direct information. When $\kappa_k = 1$, the model collapses to the standard specification with only characteristic-specific treatment effects. In the main text we treat this signal model as one interpretation that motivates allowing a common scale shift; the paper's identification and counterfactual analysis proceed directly in terms of the effective parameters and do not require separately estimating the latent signal process.

A4. Estimation Specifics

We take the published baseline demand estimates (θ_1, θ_2) and recovered mean utilities δ from Neilson (2026) as fixed inputs rather than re-estimating them in this paper. For completeness, we summarize the MPEC problem used to obtain those inputs. This formulation treats δ as an optimization variable and exploits the sparsity of the Jacobian of the market-share equations. Let $m_{\text{micro}}(\delta, \theta_2)$ denote the vector of micro moments from school choice and let $m_{\text{IV}}(\delta, \theta_1, \theta_2)$ denote the vector of IV orthogonality conditions for the demand residual. Neilson's first-step problem can be written as

$$(\theta_1^*, \theta_2^*, \delta^*) = \arg \min_{\theta_1, \theta_2, \delta} \begin{bmatrix} m_{\text{micro}}(\delta, \theta_2) - \bar{m}_{\text{micro}} \\ m_{\text{IV}}(\delta, \theta_1, \theta_2) \end{bmatrix}' \begin{bmatrix} W_{MM} & 0 \\ 0 & W_{IV-D} \end{bmatrix} \begin{bmatrix} m_{\text{micro}}(\delta, \theta_2) - \bar{m}_{\text{micro}} \\ m_{\text{IV}}(\delta, \theta_1, \theta_2) \end{bmatrix}. \quad (21)$$

The constraints are:

$$\begin{aligned} \delta &= s^{-1}(\bar{S}, \theta_2) && \text{(BLP inversion matching observed shares)} \\ \xi_{jt}(\delta, \theta_1, \theta_2) &= \delta_{jt} - f_{jt}(\theta_1) && \text{(demand residual)} \\ \xi^{\text{norm}} &= 0 && \text{(normalization restrictions),} \end{aligned}$$

where $f_{jt}(\theta_1) = \sum_r \eta_k^r x_{jt}^r$ is the linear index in observed school characteristics. The normalization restriction means that, within each market-year, mean utilities are defined relative to

one designated normalization school, so δ_{jt} and ξ_{jt} are identified only up to a market-year-specific additive constant. Thus, $m_{\text{micro}}(\delta, \theta_2)$ corresponds to model-implied averages of the chosen school characteristics by family type, $\bar{m}_{\text{micro}}$ is the corresponding vector in the data, and $m_{\text{IV}}(\delta, \theta_1, \theta_2)$ stacks the sample analog of $\mathbb{E}[Z_{jt}\xi_{jt}] = 0$ for the excluded demand instruments Z_{jt} .

A4.1. Demand Moments and Excluded Instruments

We adopt the baseline demand system from Neilson (2026). For convenience, we summarize the moments and excluded instruments used in Neilson’s estimation, since they are central to the interpretation of our treatment and equilibrium exercises.

Moments. Neilson’s demand estimation combines three types of moments. First, *market share moments* match observed school-level shares and recover mean utilities through the BLP inversion. Second, *micro moments* match the average quality, out-of-pocket payment, and distance chosen by each family type in the micro-data. Third, *IV moments* impose orthogonality between the demand residual and excluded instruments. These moments play different roles. The market-share moments recover the school-level unobserved demand component, the micro moments discipline heterogeneity in preferences across family types, and the IV moments address the endogeneity of price and academic quality. Thus the micro moments complement, rather than replace, the market-share inversion and IV conditions. In Neilson (2026), school-level choice data are available for the full population in the modeled markets, while family-specific distance information is available for a smaller but still very large geocoded subsample; accordingly, the distance micro moments are constructed from that subsample, whereas the market-share and IV moments use the broader administrative panel. For our experimental second step, the distance moment is likewise built from family-specific home-to-chosen-school driving distance, while the counterfactual simulations use node-to-school driving distances for the full choice set once each family is mapped to a residential location (node) within a market.

Why instruments are needed. Price and academic quality are endogenous school-side choice variables: schools may choose them in response to unobserved demand components. Following Neilson (2026), let the estimated demand residual be $\tilde{\xi}_{jt} = \xi_j + \Delta\xi_{jt}$, where ξ_j is a fixed non-academic school attribute and $\Delta\xi_{jt}$ is a period-specific shock. The key maintained conditions are that ξ_j is orthogonal to the instruments and that $\Delta\xi_{jt}$ is independent of both the instruments and the school’s price and quality choices. Under these assumptions, excluded instruments shift

school incentives without directly moving household demand through unobserved school appeal.

Excluded demand instruments. The main excluded instruments are:

- *Policy-exposure instrument:* the timing of the SEP reform interacted with a school's exposure to SEP-eligible students in its surrounding neighborhood.
- *Simulated transfer instrument:* the change in government transfers implied by the SEP reform when school composition is held fixed at its pre-policy level.
- *Local labor-cost shifter:* a location-by-year proxy for the cost of hiring skilled workers, constructed from wage residuals outside the school sector.

The first two instruments primarily shift schools' revenue and pricing incentives. The local labor-cost shifter is the most direct excluded source of variation for academic quality, because quality provision in this setting is teacher-intensive and labor spending accounts for the large majority of school costs. Together, these instruments provide excluded variation for the endogenous price and quality choices in the baseline demand system.

In the second step, we estimate the treatment-effect parameter vector θ_2^T by matching treatment effects in the experimental data to treatment effects simulated by the model. Let

$$m_{\text{RCT}}(\theta_2^T)$$

denote the vector of second-step moment residuals. In the specification with separate scale parameters for SEP and non-SEP families, this vector stacks the twelve moments described in Section 5:

- treatment effects on chosen quality, sticker price, out-of-pocket payment, and distance, each by SEP status;
- treatment effects on voucher-school attendance, by SEP status; and
- treatment effects on residualized mean utility $\tilde{\delta}_j \equiv \delta_j - \bar{\beta}^q q_j$, by SEP status.

Each element of $m_{\text{RCT}}(\theta_2^T)$ is therefore a difference between an experimental treatment coefficient estimated in the data, $\hat{\beta}^{\text{RCT}}$, and the analogous coefficient computed from simulated choices, $\hat{\beta}^{\text{sim}}(\theta_2^T)$. The second-step problem is

$$\theta_2^{T*} = \arg \min_{\theta_2^T} m_{\text{RCT}}(\theta_2^T)' W_{\text{RCT}} m_{\text{RCT}}(\theta_2^T). \quad (22)$$

In the common- κ specification, the $\tilde{\delta}$ moments are omitted and $m_{\text{RCT}}(\theta_2^T)$ contains ten moments instead of twelve.

In the third step, we estimate supply-side parameters (θ_3) from the quality first-order conditions. For an unconstrained school-year, the implemented residual is

$$\Omega_{jt}^U(\theta_3) = \text{CompQ}_{jt} - \frac{R_{jt} - \gamma_{\text{SEP}} \text{SEP}_{jt} - \gamma_{mt}^{\text{type}(j)}}{c_j + \delta_{\text{type}(j)}(t - 2006)}, \quad \text{CompQ}_{jt} \equiv q_{jt} + s_{jt} \left[\frac{\partial s_{jt}}{\partial q_{jt}} \right]^{-1}. \quad (23)$$

For a capacity-constrained school-year, the residual instead imposes the composition-channel condition:

$$\Omega_{jt}^C(\theta_3) = \Delta R_{jt} \frac{\partial s_{jt}^{\text{SEP}} / \partial q_{jt}}{s_{jt}} - [c_j + \delta_{\text{type}(j)}(t - 2006)].$$

The estimation uses Ω_{jt}^U or Ω_{jt}^C according to the capacity classification described in Section 5.5.5. It concentrates out the school-specific quality cost c_j and the market-year-type effect $\gamma_{mt}^{\text{type}(j)}$ jointly by alternating projections. The freely estimated parameters are γ_{SEP} , δ_{voucher} , and δ_{private} .

Let Z_{jt}^S denote the supply instrument matrix. The supply moments are the sample analog of

$$\mathbb{E}[Z_{jt}^S \Omega_{jt}(\theta_3)] = 0,$$

and we stack them in a vector $m_S(\theta_3)$. The matrix contains the baseline demand instruments and predetermined school characteristics, interactions of that matrix with for-profit and nonprofit-voucher indicators, and a new cost-side instrument: the inverse-distance-weighted average of public-school teacher PSU scores in the same market-year. The excluded baseline instruments are simulated SEP transfer changes, reform timing interacted with pre-policy neighborhood SEP exposure, and local non-education earnings deviations. SEP participation itself is the structural cost shifter whose coefficient is estimated; it is not the excluded instrument. The third-step GMM problem is

$$\theta_3^* = \arg \min_{\theta_3} m_S(\theta_3)' W_{IV-S} m_S(\theta_3). \quad (24)$$

A4.2. Computational Details

Numerical integration. The demand model includes a single random coefficient on quality ($\nu_i \sim N(0, \sigma^2)$). All integration over this random coefficient—in the market share formula (Equation 2), in the simulated treatment effects (Equation 12), and in the supply-side markdowns—uses 5-point Gauss-Hermite quadrature, a deterministic rule that is exact for polynomials up to degree 9. No Monte Carlo simulation draws are used in the estimation.

Weighting matrices. For Step 1, we use the published demand weighting matrix from Neilson (2026):

$$W = \text{diag}(W_{MM}, W_{IV-D}).$$

The second-step (treatment effects) and third-step (supply) estimations each use two-step efficient GMM: the first GMM step uses the identity matrix $W^{(1)} = (1/n_m) \cdot I$, and the second uses the inverse of the estimated covariance matrix of the moments: $W_{\text{RCT}}^{(2)} = \hat{V}_{\text{CR}}^{-1}$ (clustering at the preschool center level) for Step 2, and $W_S^{(2)} = \hat{V}_S^{-1}$ for Step 3.

Optimization algorithms and initialization. In Step 1, our code loads the published baseline demand estimates from Neilson (2026) rather than re-estimating them. Step 2 uses a constrained optimization algorithm with analytical gradients, a nonnegativity bound on the SEP quality-shift parameter, bounds $\kappa \in [0.5, 4.0]$, and a multi-start strategy. In the preferred two- κ specification reported in the paper, we use 7 starting points in each of the two GMM steps; all 7 converge to the same objective value and the same parameter estimates in the identity-weighted step, and all 7 converge to the same objective value and the same parameter estimates in the efficient-GMM step. The optimization tolerances are 10^{-8} (optimality) and 10^{-10} (step size). Step 3 concentrates out the school-specific quality costs c_j and market-year-type effects $\gamma_{mt}^{\text{type}(j)}$ via an alternating-projection algorithm. The three free parameters are γ_{SEP} , δ_{voucher} , and δ_{private} . The concentrated objective is minimized using a Nelder-Mead simplex algorithm with tolerances of 10^{-12} (function) and 10^{-10} (parameter).

A5. Calculating Standard Errors

The standard errors of the estimated parameters in each step of the estimation procedure are obtained from the variance-covariance matrix for the GMM estimator proposed by Hansen (1982). We will discuss how we calculate the standard error for a generic case (the parameters

θ and moments $\mathbf{M}$), and then we discuss the case of each set of parameters more specifically. Each one of our GMM estimators is the result of an optimization problem in which the objective function has a quadratic form:

$$\min_{\theta} Q_{obj} = \mathbf{M}' W_m \mathbf{M}$$

For which the gradient is:

$$\frac{\partial Q_{obj}}{\partial \theta} = 2 \cdot \mathbf{J}'_M W_m \mathbf{M}$$

Where $\mathbf{J}_M$ is the Jacobian.

The variance-covariance matrix for a GMM estimator is calculated using the estimator proposed by Hansen (1982):

$$\text{cov}(\theta) = (\mathbf{J}'_M W_m \mathbf{J}_M)^{-1} \mathbf{J}'_M W_m \mathbf{V} W'_m \mathbf{J}_M (\mathbf{J}'_M W_m \mathbf{J}_M)^{-1}$$

Where V is the variance-covariance matrix of the moments. With deterministic Gauss-Hermite quadrature, there is no simulation error in the model predictions. For the parameters estimated from the experimental moments, the relevant source of uncertainty is therefore sampling error in the treatment-effect estimates. As discussed by Berry et al. (2004), when both sources are present they enter additively; in our current implementation only the sampling component remains. The RCT moments in Equation 12 take the difference between the estimated treatment effects and the model's predictions, so the variance of those moments is estimated from the cluster-robust variance-covariance matrix of the underlying treatment-effect regressions.